

NEET 2025 - Paper Code - Narmada 48

Physics

Question 1: A parallel plate capacitor made of circular plates is being charged such that the surface charge density on its plates is increasing at a constant rate with time. The magnetic field arising due to displacement current is:

Options:

- (a) Non-zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates
- (b) Zero between the plates and non-zero outside
- (c) Zero at all places
- (d) Constant between the plates and zero outside the plates

Answer: (a)

Solution: Non-zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_c + \epsilon_0 \frac{\partial \phi}{\partial t} \right)$$

$\therefore$ Maxwell Ampere law

In the gap, there's no conduction but there is displacement current due to changing electric field.

This current leads to magnetic field in the region

If increases radially with in the places, reaching maximum at the edge.

Question 2: An electric dipole with dipole moment $5 \times 10^{-6} \text{ Cm}$ is aligned with the direction of a uniform electric field of magnitude $4 \times 10^5 \text{ N/C}$. The dipole is then rotated through an angle of 60° with respect to the electric field. The change in the potential energy of the dipole is:

Options:

- (a) 1.2 J
- (b) 1.5 J
- (c) 0.8 J
- (d) 1.0 J

Answer: (d)

Solution: $\Delta U = U_2 - U_1$

$$= (-pE \cos \theta_2) - (-pE \cos \theta_1)$$

$$= -pE \cos 60 + pE \cos 0$$

$$= \frac{-pE}{2} + pE$$

$$= \frac{1}{2} pE = \frac{1}{2} \times 5 \times 10^{-6} \times 4 \times 10^5 = 1.0 \text{ J}$$

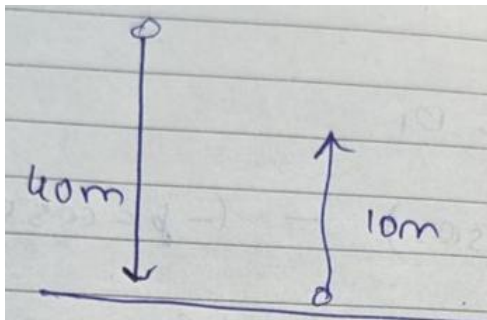
Question 3: A ball of mass 0.5 kg is dropped from a height of 40 m. The ball hits the ground and rises to a height of 10 m. The impulse imparted to the ball during its collision with the ground is (Take $g = 9.8 \text{ m/s}^2$)

Options:

- (a) 0
- (b) 84 NS
- (c) 21 NS
- (d) 7 NS

Answer: (c)

Solution:



$$v_1 = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 40} = \sqrt{196 \times 4}$$

$$= 14 \times 2 = 28 \text{ m/s}$$

$$v_2 = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 10} = \sqrt{196} = 14 \text{ m/s}$$

$$3 = p_2 - p_1$$

$$= mv_2 - mv_1$$

$$= 0.5(+14) - 0.5(-28) = 7 + 14 = 21$$

Question 4: The intensity of transmitted light when a polaroid sheet, placed between two crossed polaroids at 22.5° from the polarization axis of one of the polaroid, is (I_0 is the intensity of polarized light after passing through the first polaroid):

Options:

(a) $\frac{I_0}{8}$

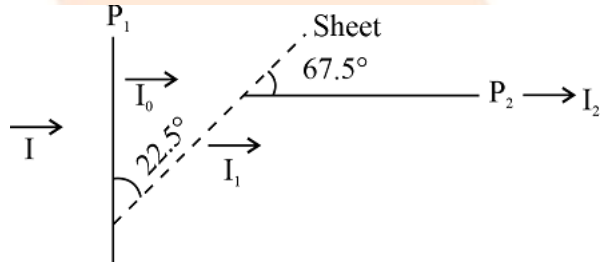
(b) $\frac{I_0}{16}$

(c) $\frac{I_0}{2}$

(d) $\frac{I_0}{4}$

Answer: (a)

Solution:



$$I_1 = I_0 \cos^2 \theta = I_0 \cos^2 (22.5)$$

$$I_2 = I_1 \cos^2 \theta = I_1 \cos^2 67.5$$

$$I_2 = (I_0 \cos^2 22.5) (\cos^2 67.5) \dots (1)$$

$$\cos 22.5 = \sqrt{\frac{1 + \cos 45}{2}} = \sqrt{1 + \frac{1}{\sqrt{2}}} = \sqrt{\frac{1}{2} + \frac{1}{2\sqrt{2}}}$$

$$\cos 67.5 = \sin 22.5 = \sqrt{\frac{1 - \cos 45}{2}} = \sqrt{\frac{1}{2} - \frac{1}{2\sqrt{2}}}$$

From (1)

$$I_2 = I_0 \left(\frac{1}{2} + \frac{1}{2\sqrt{2}} \right) \left(\frac{1}{2} - \frac{1}{2\sqrt{2}} \right)$$

$$= I_0 \left(\frac{1}{4} - \frac{1}{4 \times 2} \right) = I_0 \left(\frac{1}{4} - \frac{1}{8} \right)$$

$$= \frac{I_0}{8}$$

Question 5: The kinetic energies of two similar cars A and B are 100 J and 225 J respectively. On applying breaks, car A stops after 1000 m and car B stops after 1500 m. If F_A and F_B are the forces applied by the breaks on cars A and B respectively, then the ratio $\frac{F_A}{F_B}$ is

Options:

- (a) $\frac{1}{3}$
- (b) $\frac{1}{2}$
- (c) $\frac{3}{2}$
- (d) $\frac{2}{3}$

Answer: (d)

Solution: $W = \Delta KE$

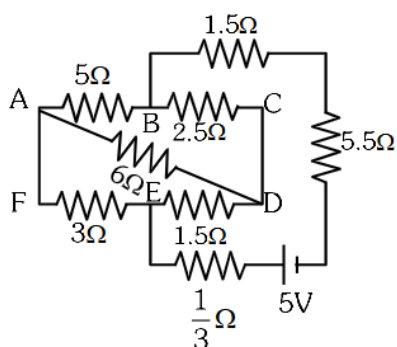
$$-Fs = 0 - KE$$

$$Fs = KE$$

$$F_1 = \frac{100}{1000} = \frac{1}{10}, F_2 = \frac{225}{1500} = \frac{15}{100}$$

$$\frac{F_1}{F_2} = \frac{1}{10} \times \frac{100}{15} = \frac{10}{15} = \frac{2}{3}$$

Question 6: The current passing through the battery in the given circuit, is:



Options:

- (a) 2.5 A
- (b) 1.5 A
- (c) 2.0 A
- (d) 0.5 A

Answer: (d)

Solution: ABCDEF forms wheat stone $5 \times 1.5 = 7.5$

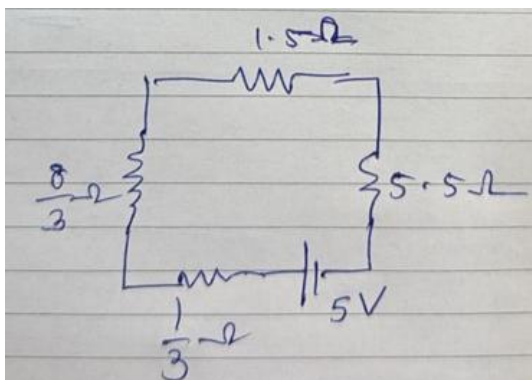
$$3 \times 2.5 = 7.5$$

$\therefore$ Ignoring 6Ω resistance $3\Omega, 5\Omega$ in series gives 4Ω

$1.5, 2.5$ in series gives 4Ω

Now, $8\Omega, 4\Omega$ in parallel gives $\frac{8 \times 4}{8 + 4} = \frac{8}{3}\Omega$

New circuit:

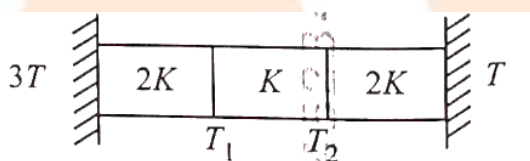


$$V = IR = i \left(\frac{1}{3} + \frac{8}{3} + 1.5 + 5.5 \right) = i(10)$$

$$5 = i(10)$$

$$i = 0.5A$$

Question 7: Three identical heat conducting rods are connected in series as shown in the figure. The rods on the sides have thermal conductivity $2K$ while that in the middle has thermal conductivity K . The left end of the combination is maintained at temperature $3T$ and the right end at T . The rods are thermally insulated from outside. In steady state, temperature at the left junction is T_1 and that at the right junction is T_2 . The ratio $\frac{T_1}{T_2}$ is

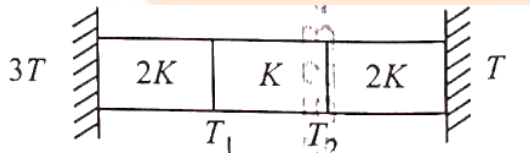


Options:

- (a) $\frac{5}{3}$
- (b) $\frac{5}{4}$
- (c) $\frac{3}{2}$
- (d) $\frac{4}{3}$

Answer: (a)

Solution:



$$Q = \frac{kA\Delta T}{L}$$

$$Q = \frac{2kA}{L}(3T - T_1) = \frac{kA}{L}(T_1 - T_2) = \frac{2kA}{L}(T_2 - T)$$

$$\therefore 2(3T - T_1) = (T_1 - T_2) = 2(T_2 - T) = \text{constant}$$

$$\therefore \boxed{6T - 2T_1 = C} \dots (1)$$

$$\boxed{T_1 - T_2 = C} \dots (2)$$

$$\boxed{2T_2 - T = C} \dots (3)$$

$$\boxed{2T_1 - 2T_2 = 2C} \dots (4)$$

And (1) & (4)

$$6T - 2T_2 = 3C \quad \text{..(5)}$$

$$2T_2 - 2T = C \quad \text{from (3)}$$

Add above $4T = 4C$

$$T = C$$

From (1), $6T - 2T_1 = C$

$$6C - 2T_1 = C$$

$$5C = 2T_1$$

$$\frac{5C}{2} = T_1$$

From 3, $2T_2 - 2T = C$

$$2T_2 - 2C = C$$

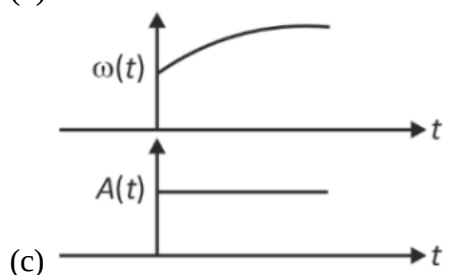
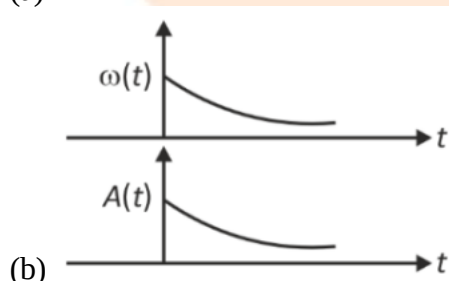
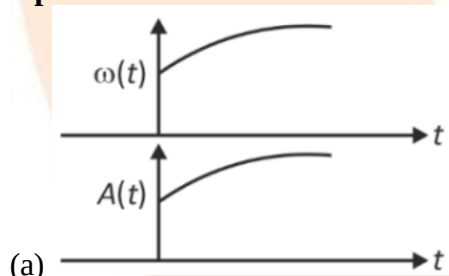
$$2T_2 = 3C$$

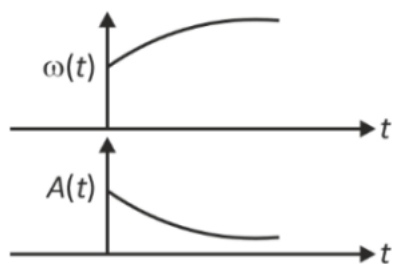
$$T_2 = \frac{3C}{2}$$

$$\frac{T_1}{T_2} = \frac{\frac{5C}{2}}{\frac{3C}{2}} = \frac{5}{3}$$

Question 8: In an oscillating spring mass system, a spring is connected to a box filled with sand. As the box oscillates, sand leaks slowly out of the box vertically so that the average frequency $\omega(t)$ and average amplitude $A(t)$ of the system change with the time t . Which one of the following options schematically depicts these changes correctly?

Options:





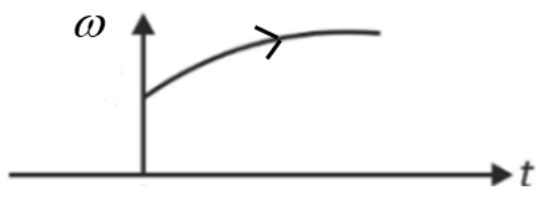
(d)

Answer: (d)

Solution:

$$\omega = \sqrt{\frac{k}{m}}$$

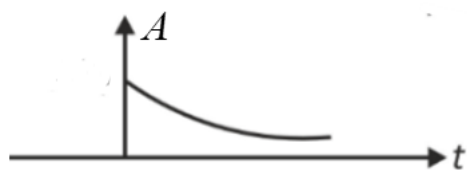
$m \downarrow, \omega \uparrow$ until it becomes constant



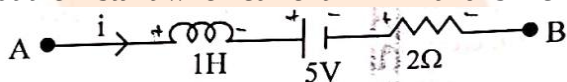
$$E = \frac{1}{2} k A^2$$

$$\sqrt{\frac{2E}{k}} = A$$

If energy is lost due to leakage, amplitude can decrease.



Question 9: AB is a part of an electrical circuit (see figure). The potential difference " $V_A - V_B$ ", at the instant when current $i = 2A$ and is increasing at a rate of 1 amp/second is:

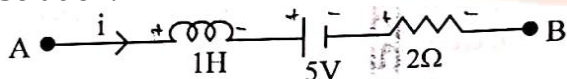


Options:

- (a) 9 volt
- (b) 10 volt
- (c) 5 volt
- (d) 6 volt

Answer: (b)

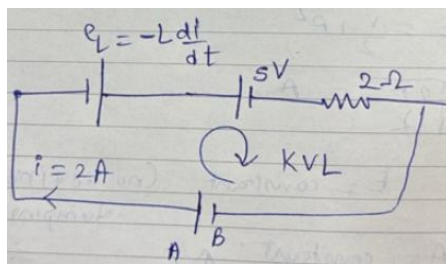
Solution:



$$\frac{di}{dt} = + \frac{1A}{s}$$

Redraw

$$e_L = -L \frac{di}{dt}$$



$$(V_A - V_B) + e_L - 5 - iR = 0$$

$$(V_A - V_B) - L \frac{di}{dt} - 5 - iR = 0$$

$$(V_A - V_B) - 1(1) - 5 - 2(2) = 0$$

$$V_A - V_B = 10V$$

Question 10: A particle of mass m is moving around the origin with a constant force F pulling it towards the origin. If Bohr model is used to describe its motion, the radius of the n th orbit and the particle's speed v in the orbit depend on n as

Options:

(a) $r \propto n^{\frac{2}{3}}; v \propto n^{\frac{1}{3}}$

(b) $r \propto n^{\frac{4}{3}}; v \propto n^{-\frac{1}{3}}$

(c) $r \propto n^{\frac{1}{3}}; v \propto n^{\frac{1}{3}}$

(d) $r \propto n^{\frac{1}{3}}; v \propto n^{\frac{2}{3}}$

Answer: (a)

Solution: Bohr's quantization

$$(1) L = mvr = nh, v = \frac{nh}{mr}$$

$$(2) F = \text{constant}$$

$$\frac{mv^2}{r} = F$$

$$\frac{m}{r} \left(\frac{n^2 h^2}{m^2 r^2} \right) = F$$

$$n^2 \propto r^3$$

$$n^{\frac{2}{3}} \propto r$$

$$\therefore v = \frac{nh}{mr}$$

$$v \propto \frac{n}{r} \propto \frac{n}{n^{\frac{2}{3}}} \propto n^{\frac{1}{3}}$$

Question 11: In some appropriate units, time (t) and position (x) relation of a moving particle is given by $t = x^2 + x$. The acceleration of the particle is

Options:

(a) $+\frac{2}{(x+1)^3}$

(b) $+\frac{2}{2x+1}$

$$(c) -\frac{2}{(x+2)^3}$$

$$(d) -\frac{2}{(2x+1)^3}$$

Answer: (d)

Solution: $t = x^2 + x$

Derivative w.r.t x

$$\frac{dt}{dx} = 2x + 1$$

$$\therefore \frac{dx}{dt} = \frac{1}{2x+1}$$

$$\therefore \boxed{v = \frac{1}{2x+1}}$$

Derivative w.r.t. time t

$$\therefore a = \frac{-1}{(2x+1)^2} \frac{2dx}{dt}$$

$$= \frac{-1}{(2x+1)^2} 2v$$

$$= \frac{-1}{(2x+1)^2} 2 \left(\frac{1}{2x+1} \right)$$

$$= \frac{-2}{(2x+1)^3}$$

Question 12: A model for quantized motion of an electron in a uniform magnetic field B states that the flux passing through the orbit of the electron is $n \left(\frac{h}{e} \right)$ where n is an integer, h is

Planck's constant and e is the magnitude of electron's charge. According to the model, the magnetic moment of an electron in its lowest energy state will be (m is the mass of the electron)

Options:

(a) $\frac{heB}{\pi m}$

(b) $\frac{heB}{2\pi m}$

(c) $\frac{he}{\pi m}$

(d) $\frac{he}{2\pi m}$

Answer: (d)

Solution: $\boxed{\phi = \frac{nh}{e}}$

Given but $\phi = BA$

$$\therefore \frac{\phi}{B} = A$$

$$\therefore A = \frac{\phi}{B} = \frac{nh}{eB}$$

For $n = 1$ (lowest energy given)

$$A = \frac{h}{eB}$$

$$\pi r^2 = \frac{h}{eB}$$

$$r^2 = \frac{h}{\pi eB}$$

We know, magnetic moment $\mu = \frac{evr}{2}$

We need velocity.

Centripetal force = Magnetic force

$$\frac{mv^2}{r} = evB$$

$$\mu = \frac{eBr}{m}$$

$$\therefore \mu = \frac{evr}{2}$$

$$= \frac{e}{2} \frac{eBr}{m}$$

$$= \frac{e^2 B}{2m} r^2 = \frac{e^2 B}{2m} \frac{h}{\pi eB}$$

$$= \frac{he}{2\pi m}$$

Question 13: A microscope has an objective of focal length 2 cm, eyepiece of focal length 4 cm and the tube length of 40 cm. If the distance of distinct vision of eye is 25 cm, the magnification in the microscope is

Options:

- (a) 150
- (b) 250
- (c) 100
- (d) 125

Answer: (d)

Solution: It is a compound microscope

$$M_{\text{objective}} \approx \frac{L}{f_o} \approx \frac{40\text{cm}}{2\text{cm}} \approx 20$$

Case 1: For relaxed eye ($V = \infty$)

$$MP_{\text{eyepiece}} = \frac{D}{f_e} = \frac{25}{4} = 6.25$$

$$M_{\text{compound}} = 20 \times 6.25$$

(For relaxed eye $V = \infty$)

Case 2: For LDDV

$$MP_{\text{eyepiece}} = 1 + \frac{D}{f_e} = 7.25$$

$$M_{\text{compound}} = 20 \times 7.25 = 145$$

Note: They have NOT asked for maximum magnification explicitly.

Question 14: There are two inclined surfaces of equal length (L) and same angle of inclination 45° with the horizontal. One of them is rough and the other is perfectly smooth. A given body takes 2 times as much time to slide down on rough surface than on the smooth surface. The

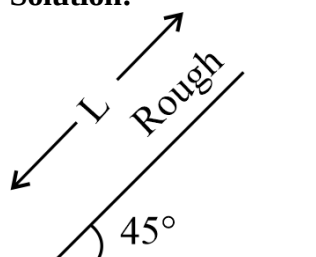
coefficient of kinetic friction (μ_k) between the object and the rough surface is close to

Options:

- (a) 0.5
- (b) 0.75
- (c) 0.25
- (d) 0.40

Answer: (b)

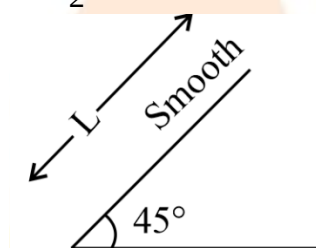
Solution:



$$t_1 = 2t$$

$$a_1 = g \sin \theta - \mu g \cos \theta$$

$$L = \frac{1}{2} a_1 t_1^2$$



$$t_2 = t$$

$$a_2 = g \sin \theta$$

$$L = \frac{1}{2} a_2 t_2^2$$

$$a_1 t_1^2 = a_2 t_2^2$$

$$(g \sin \theta - \mu g \cos \theta) 4t^2 = (g \sin \theta) t^2$$

$$4g \sin \theta - 4\mu g \cos \theta = g \sin \theta$$

$$3g \sin \theta = 4\mu g \cos \theta$$

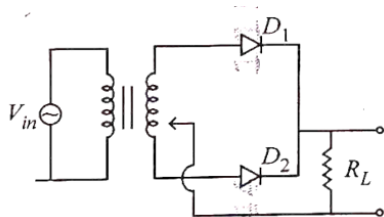
$$\tan \theta = \frac{4\mu}{3}$$

$$\tan 45 = \frac{4\mu}{3}$$

$$1 = \frac{4\mu}{3}$$

$$\mu = \frac{3}{4}$$

Question 15: A full wave rectifier circuit with diodes (D_1) and (D_2) is shown in the figure. If input supply voltage $V_{in} = 220 \sin(100\pi t)$ volt, then at $t = 15$ m sec



Options:

- (a) D_1 and D_2 both are forward biased
- (b) D_1 and D_2 both are reverse biased
- (c) D_1 is forward biased, D_2 is reverse biased
- (d) D_1 is reverse biased, D_2 is forward biased

Answer: (d)

Solution: $V_{in} = 220 \sin(100\pi t)$

$$\omega = 100\pi$$

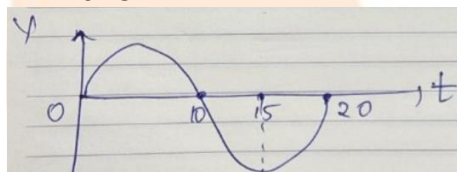
$$\frac{2\pi}{T} = 100\pi$$

$$\frac{2}{T} = 100$$

$$\frac{2}{100} = T$$

$$T = \frac{20}{1000} = s$$

$$T = 20ms$$



$\therefore$ At 15 ms

$V = -ve \text{ max}$

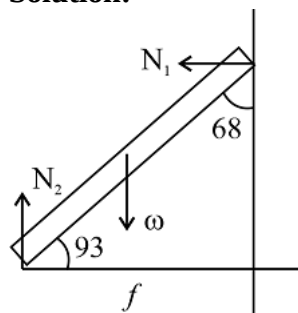
Question 16: A uniform rod of mass 20 kg and length 5 m leans against a smooth vertical wall making an angle of 60° with it. The other end rests on a rough horizontal floor. The friction force that the floor exerts on the rod is (take $g = 10 \text{ m/s}^2$)

Options:

- (a) 200 N
- (b) $200\sqrt{3} \text{ N}$
- (c) 100 N
- (d) $100\sqrt{3} \text{ N}$

Answer: (d)

Solution:



Mass $m = 20 \text{ kg}$

Length = 5m

$$w = mg = 20 \times 10 = 200N$$

The rod is in equilibrium

E_{net} on horizontal

$$N_1 = f$$

$$w = N_2$$

Since the rod is in rotational equilibrium

$$\tau_1 = \tau_2$$

$$w \times \frac{l}{2} \cos 80 = N_1 \times \frac{l}{2} \cos 60$$

$$100 \times \frac{5}{2} \times \frac{\sqrt{3}}{2} = N_1 \times 5 \times \frac{1}{2}$$

$$N_1 = 100\sqrt{3}$$

$$f = 100\sqrt{3}$$

Question 17: Two identical charged conducting spheres A and B have their centres separated by a certain distance. Charge on each sphere is q and the force of repulsion between them is F . A third identical uncharged conducting sphere is brought in contact with sphere A first and then with B and finally removed from both. New force of repulsion between spheres A and B (Radii of A and B are negligible compared to the distance of separation so that for calculating force between them they can be considered as point charges) is best given as :

Options:

(a) $\frac{F}{2}$

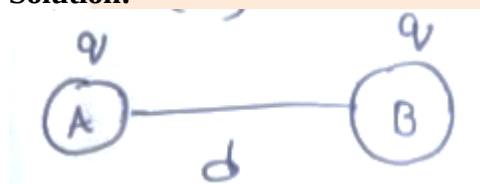
(b) $\frac{3F}{8}$

(c) $\frac{3F}{5}$

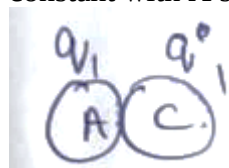
(d) $\frac{2F}{3}$

Answer: (b)

Solution:

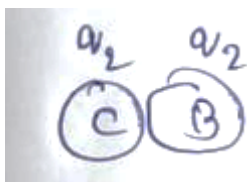


Force of repulsion $F = \frac{kq^2}{d^2}$ a third identical uncharged conducting sphere is brought in constant with A sphere and then with B.



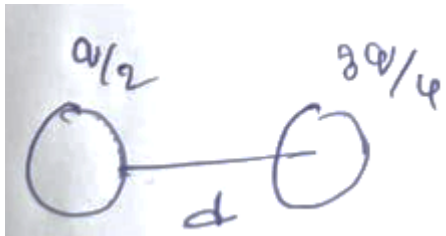
$$q_1 = \frac{q+0}{2} = \frac{q}{2}$$

Now charge on both A and C $= \frac{q}{2}$



$$q_2 = \frac{q + \frac{q}{2}}{2} = \frac{8q}{4}$$

Now force between A and B



$$F' = \frac{kq_1q_2}{d^2} = \frac{k\left(\frac{q}{2}\right)\left(\frac{8q}{4}\right)}{d^2}$$

$$F' = \frac{3}{8} \frac{kq^2}{d^2} = \frac{3}{8} F$$

Question 18: Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scotty with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

Options:

- (a) 10 min, 90 km/h
- (b) 15 min, 120 km/h
- (c) 9 min, 40 km/h
- (d) 25 min, 100 km/h

Answer: (b)

Solution: Speed of the bus = V km / h

While the girl is driving the scotty with speed 60 km/hr

From X to Y direction bus goes past her every 30 minutes in the direction of her motion 10 minutes in the opposite direction.

Bus leaves at every T min w.r.t of ground distance between two buses

$$d = VT = V \times \frac{T}{60} \text{ km} \dots (1)$$

w.r.t of max (X to Y) in same direction $V_{B/m} = (V - 60)$

$$\text{Distance } D = (V - 60) \frac{30}{60} \dots (2)$$

In opposite direction $V_{B/m} = v + 60$

$$D = (V + 60) \frac{10}{60} \dots (3)$$

$$(1) = (3)$$

$$(V - 60) \frac{30}{60} = (V + 60) \frac{10}{60}$$

$$3V - 180 = V + 60$$

$$2V = 240$$

$$V = 120 \text{ km / hr}$$

$$(1) = (2)$$

$$\frac{VT}{60} = \frac{(V - 60)30}{60}$$

$$120 \times T = (20 - 60)30$$

$$120 \times T = 60 \times 30$$

$$T = 15 \text{ min}$$

Question 19: A container has two chambers of volumes $V_1 = 2$ litres and $V_2 = 3$ litres separated by a partition made of a thermal insulator. The chambers contains $n_1 = 5$ and $n_2 = 4$ moles of ideal gas at pressure $p_1 = 1$ atm and $p_2 = 2$ atm, respectively. When the partition is removed, the mixture attains an equilibrium pressure of:

Options:

(a) 1.4 atm

(b) 1.8 atm

(c) 1.3 atm

(d) 1.6 atm

Answer: (d)

Solution: $v_1 = 2 \text{ Lt}, v_2 = 3 \text{ Lt}$

$$n_1 = 5 \text{ mol}, n_2 = 4 \text{ mol}$$

$$p_1 = 1 \text{ atm}, p_2 = 2 \text{ atm}$$

$$\text{Chamber 1 } P_1 V_1 = n_1 R T_1$$

$$\text{Chamber 2 } P_2 V_2 = n_2 R T_2$$

When partition is removed

Gasses will mix without loss of energy

$$\text{From kinetic theory of gasses } PV = \frac{2}{3} n (K.E)$$

$$\text{For chamber 1 } P_1 V_1 = \frac{2}{3} n_1 E_1$$

$$\text{For Chamber 2 } P_2 V_2 = \frac{2}{3} n_2 E_2$$

$$\frac{3}{2} P_1 V_1 = n_1 E_1$$

$$\Rightarrow (n_1 E_1 + n_2 E_2) = \frac{3}{2} (P_1 V_1 + P_2 V_2)$$

$$\frac{3}{2} P_2 V_2 = n_2 E_2$$

$$\text{We know } PV = \frac{2}{3} nE$$

$$P(V_1 + V_2) = [P_1 V_1 + P_2 V_2]$$

$$P = \frac{P_1 V_1 + P_2 V_2}{V_1 + V_2} = \frac{2 \times 1 + 2 \times 3}{2 + 3}$$

$$P = \frac{2 + 6}{5} = \frac{8}{5} = 1.6$$

Question 20: De-Broglie wavelength of an electron obtaining in the $n = 2$ state of hydrogen atom is close to (Given Bohr radius = 0.052 nm)

Options:

(a) 1.67 nm

- (b) 2.67 nm
(c) 0.067 nm
(d) 0.67 nm

Answer: (d)

Solution: $n = 2$

De Broglie wavelength $\lambda = ?$

$$r = 0.052 \text{ nm}$$

$$\lambda = \frac{2\pi r}{n}$$

$$r = n^2 \cdot r_0 = 2^2 \times 0.052$$

$$= 4 \times 0.052 = 0.208$$

$$r = 0.208$$

$$\lambda = \frac{2 \times 3.14 \times 0.208}{2} = 0.653$$

$$\cong 0.67 \text{ nm}$$

Question 21: To an ac power supply of 220V at 50 Hz, a resistor of 20Ω , a capacitor of reactance 25Ω and an inductor of reactance 45Ω are connected in series. The corresponding current in the circuit and the phase angle between the current and the voltage is, respectively -

Options:

- (a) 1.56 A and 30°
(b) 1.56 A and 45°
(c) 7.8 A and 30°
(d) 7.8 A and 45°

Answer: (d)

Solution: $V = 220 \text{ V}$, $f = 50 \text{ Hz}$

$$R = 20\Omega, X_C = 25\Omega, X_L = 45\Omega$$

$$I = \frac{V}{Z}$$

$$Z = \sqrt{R^2 + X^2}$$

$$X = X_L - X_C$$

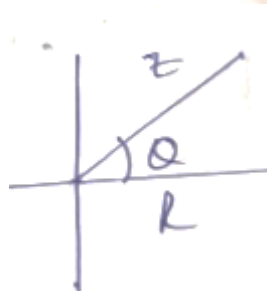
$$X = 45 - 25 = 20$$

$$Z = \sqrt{20^2 + 20^2} = \sqrt{800}$$

$$Z = 20\sqrt{2}$$

$$I = \frac{V}{Z} = \frac{220}{20\sqrt{2}} = \frac{11}{\sqrt{2}}$$

$$I = 7.8$$



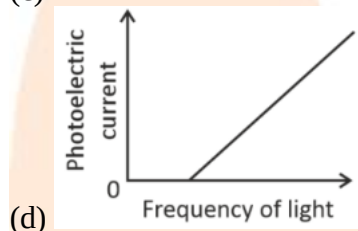
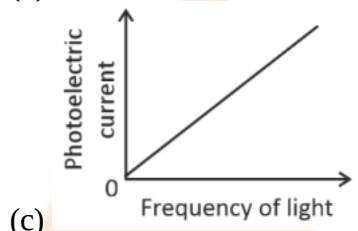
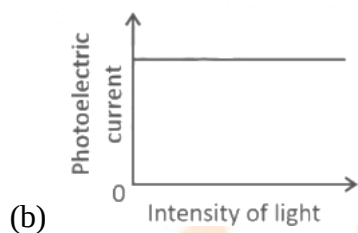
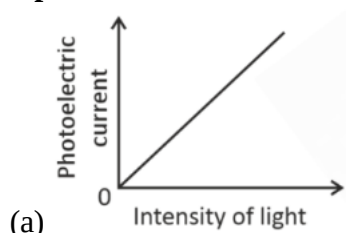
$$\cos \theta = \frac{R}{Z} = \frac{20}{20\sqrt{2}}$$

$$\cos \theta = \cos 45$$

$$\theta = 45^\circ$$

Question 22: Which of the following options represent the variation of photoelectric current with property of light shown on the x-axis?

Options:



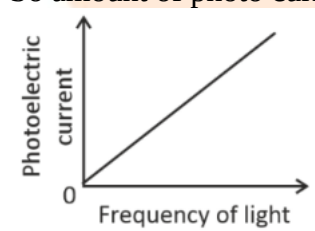
Answer: (c)

Solution: The intensity of light refers to the amount of photon energy per unit area.

So greater intensity of light more will be number of photons, as a result, the more will be the number ejected photo electron. More photo electrons constitute more photo current

Hence if intensity increased, photo current also gets increased.

So amount of photo current lineary dependent on the intensity of incident ratidation .



Question 23: A pipe open at both ends has a fundamental frequency f in air. The pipe is now dipped vertically in a water drum to half of its length. The fundamental frequency of the air column is now equal to:

Options:

(a) $\frac{3f}{2}$

(b) $2f$

(c) $\frac{f}{2}$

(d) f

Answer: (d)

Solution: Before dipping $f = \frac{nv}{2L} = \frac{v}{2L} \dots(1)$

After dipping in water $f_1 = \frac{nv}{4\left(\frac{L}{2}\right)} = \frac{v}{2L} \dots(2)$

From (1) & (2)

$$\boxed{f_1 = f}$$

Question 24: Two identical point masses P and Q, suspended from two separate massless springs of spring constant k_1 and k_2 , respectively, oscillate vertically. If their maximum speeds are the same, the ratio $\left(\frac{A_Q}{A_P}\right)$ of the amplitude A_Q of mass Q to the amplitude A_P of mass P is:

Options:

(a) $\sqrt{\frac{k_2}{k_1}}$

(b) $\sqrt{\frac{k_1}{k_2}}$

(c) $\frac{k_2}{k_1}$

(d) $\frac{k_1}{k_2}$

Answer: (b)

Solution: $A_1\omega_1 = A_2\omega_2$

$$A_P\omega_P = A_Q\omega_Q$$

$$\omega = \frac{2\pi}{T}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$\Rightarrow \frac{A_P}{A_Q} = \frac{\omega_Q}{\omega_P}$$

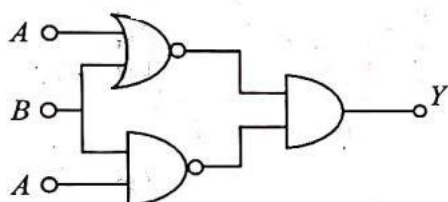
$$= \frac{T_P}{T_Q}$$

$$= \frac{\sqrt{k_Q}}{\sqrt{k_P}}$$

$$= \sqrt{k_2 : k_1}$$

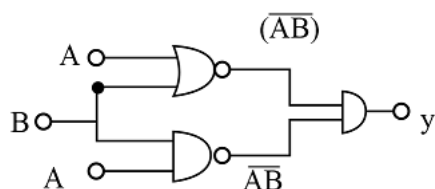
$$A_Q : A_P = \sqrt{k_1 : k_2}$$

Question 25: The output (Y) of the given logic implementation is similar to the output of an/a _____ gate.



Options:

- (a) OR
- (b) NOR
- (c) AND
- (d) NAND

Answer: (b)**Solution:**

$$\begin{aligned}
 y &= (\overline{A+B}) \overline{AB} \\
 &= (\overline{A} \cdot \overline{B}) (\overline{A+B}) \\
 &= \overline{A} \overline{B} + \overline{A} \overline{B} \overline{B} \\
 &= \overline{A} \overline{B} + \overline{A} \overline{B} \\
 &= \overline{A} \overline{B} \\
 &= \overline{A \cdot B} \\
 &= \overline{A+B} = \text{NOR}
 \end{aligned}$$

Question 26: An oxygen cylinder of volume 30 litre has 18.20 moles of oxygen. After some oxygen is withdrawn from the cylinder, its gauge pressure drops to 11 atmospheric pressure at temperatures 27°C . The mass of the oxygen withdrawn from the cylinder is nearly equal to:

[Given, $R = \frac{100}{12} \text{ J mol}^{-1} \text{ K}^{-1}$, and molecular mass of $\text{O}_2 = 32$, 1 atm pressure

$$= 1.01 \times 10^5 \text{ N/m}^2]$$

Options:

- (a) 0.116 kg
- (b) 0.156 kg
- (c) 0.125 kg
- (d) 0.144 kg

Answer: (b)**Solution:** Since $PV = nRT$

New no. of moles

$$\therefore n' = \frac{PV}{RT} = \frac{11 \times 1.01 \times 30 \times 10^{-3} \times 10^5}{\frac{100}{12} \times 300} = \frac{33330}{2500}$$

$$n' = 13.33$$

$$\text{Now } M = (n - n') \times 32$$

$$= (4.87) \times 32 \text{ g}$$

$$= 155.84 \text{ g or } 0.1558 \text{ kg}$$

$$= 0.156 \text{ kg}$$

Question 27: In a certain camera, a combination of four similar thin convex lenses are arranged axially in contact. Then the power of the combination and the total magnification in comparison to the power (p) and magnification (m) for each lens will be, respectively -

Options:

- (a) $4p$ and m^4
- (b) p^4 and m^4
- (c) $4p$ and $4m$
- (d) p^4 and $4m$

Answer: (a)

Solution: Power is additive when lenses are in contact

So each lens has power P , then 4 lenses

$$P_{\text{total}} = P + P + P + P = 4P$$

Magnification $m_{\text{total}} = m \times m \times m \times m$ (series) $= m^4$

Question 28: Two gases A and B are filled at the same pressure in separate cylinders with movable pistons of radius r_A and r_B , respectively. On supplying an equal amount of heat to both the systems reversibly under constant pressure, the pistons of gas A and B are displaced by 16 cm and 9 cm, respectively. If the change in their internal energy is the same, then the ratio

$\frac{r_A}{r_B}$ is equal to

Options:

- (a) $\frac{2}{\sqrt{3}}$
- (b) $\frac{\sqrt{3}}{2}$
- (c) $\frac{4}{3}$
- (d) $\frac{3}{4}$

Answer: (d)

Solution: ΔU are same

$\Delta\theta$ are same.

$\therefore \Delta\omega$ should be same

$$P\pi r_A^2 L_A = P\pi r_B^2 L_B$$

$$\frac{r_A}{r_B} = \sqrt{\frac{L_B}{L_A}} = \sqrt{\frac{9}{16}} = \frac{3}{4}$$

Question 29: A balloon is made of a material of surface tension S and its inflation outlet (from where gas is filled in it) has small area A . It is filled with a gas of density ρ and takes a spherical shape of radius R . When the gas is allowed to flow freely out of it, its radius r changes from R to 0 (zero) in time T . If the speed $v(r)$ of gas coming out of the balloon depends on r as r^a and $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$ then

Options:

- (a) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$
- (b) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -\frac{1}{2}, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$
- (c) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -1, \gamma = +1, \delta = \frac{3}{2}$
- (d) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = -\frac{1}{2}, \delta = \frac{5}{2}$

Answer: (a)

Solution: $T \propto S^\alpha A^\beta \rho^r R^\delta$

$$[T] = [MT^{-2}]^\alpha [L^2]^\beta [ML^{-3}]^\gamma [L^\delta]$$

$$[T] = [M^{\alpha+\gamma} L^{2\beta-3\gamma+\delta} T^{-2\alpha}]$$

Compare powers

$$-2\alpha = 1 \dots (1) \Rightarrow \alpha = -\frac{1}{2}$$

$$\alpha + \gamma = 0 \dots (2) \Rightarrow \gamma = \frac{1}{2}$$

$$2\beta - 3\gamma + \delta = 0 \dots (3)$$

$$2\beta - \frac{3}{2} + \delta = 0$$

$$2\beta = \frac{3}{2} - \delta$$

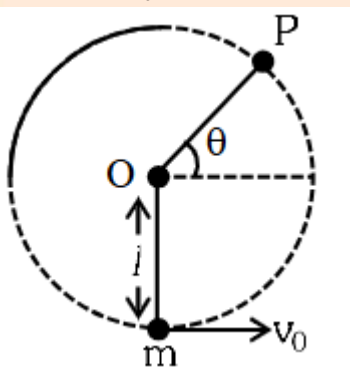
$$\text{If } \delta = \frac{7}{2}$$

$$2\beta = \frac{3}{2} - \frac{7}{2} = \frac{-4}{2} = -2$$

$$\beta = -1$$

$$\alpha = -\frac{1}{2}, \gamma = \frac{1}{2}, \beta = -1, \delta = \frac{7}{2}$$

Question 30: A bob of heavy mass m is suspended by a light string of length l . The bob is given a horizontal velocity v_0 as shown in figure. If the string gets slack at some point P making an angle θ from the horizontal, the ratio of the speed v of the bob at point P to its initial speed v_0 is:



Options:

(a) $\left(\frac{\cos \theta}{2 + 3 \sin \theta} \right)$

(b) $\left(\frac{\sin \theta}{2 + 3 \sin \theta} \right)$

(c) $(\sin \theta)^{\frac{1}{2}}$

(d) $\left(\frac{1}{2 + 3 \sin \theta} \right)^{\frac{1}{2}}$

Answer: (b)

Solution: $\frac{mv^2}{l} = mg \sin \theta$

$$\frac{1}{2}mv_0^2 = mgl(1 + \sin \theta) + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv_0^2 = mgl(1 + \sin \theta) + \frac{1}{2}mgl \sin \theta$$

$$V_0^2 = 2gl + 2gl \sin \theta + gl \sin \theta$$

$$V_0^2 = gl(2 + 3\sin \theta)$$

$$\frac{V_0^2}{V^2} = \frac{gl(2 + 3\sin \theta)}{gl \sin \theta}$$

$$\frac{V_0}{V} = \sqrt{\frac{2 + 3\sin \theta}{\sin \theta}}$$

$$\frac{V}{V_0} = \sqrt{\frac{\sin \theta}{2 + 3\sin \theta}}$$

Question 31: A physical quantity P is related to four observations a, b, c and d as follows:

$$P = a^3b^2 / c\sqrt{d}$$

The percentage errors of measurement in a, b, c and d are 1%, 3%, 2% and 4% respectively.

The percentage errors in the quantity P is

Options:

- (a) 13 %
- (b) 15 %
- (c) 10 %
- (d) 2 %

Answer: (a)

Solution: $P = a^3b^2 / c\sqrt{d}$

$$= a^3b^2c^{-1}d^{-1/2}$$

$$\% \Delta p = 3 \left(\frac{\Delta a}{a} \times 100 \right) + 2 \left(\frac{\Delta b}{b} \times 100 \right) + 1 \left(\frac{\Delta c}{c} \times 100 \right) + \frac{1}{2} \left(\frac{\Delta d}{d} \times 100 \right)$$

$$= (3 \times 1) + (2 \times 3) + (1 \times 2) + \frac{1}{2}(4)$$

$$= 3 + 6 + 2 + 2 = 13\%$$

Question 32: The Sun rotates around its centre once in 27 days. What will be the period of revolution if the Sun were to expand to twice its present radius without any external influence? Assume the Sun to be a sphere of uniform density.

Options:

- (a) 115 days
- (b) 108 days
- (c) 100 days
- (d) 105 days

Answer: (b)

Solution: External force = 0

By applying cons of angular momentum

T = Initial time period = 27 days

T' = Final time period

ω' = Final angular velocity

$I_i(\text{System}) = I_f(\text{System})$

$$\Rightarrow I\omega = I'\omega'$$

$$\Rightarrow \left(\frac{2}{5}MR^2 \right) \omega = \frac{2}{5}M(2R)^2 \omega'$$

$$\Rightarrow \omega R^2 = 4R^2 \omega'$$

$$\omega' = \frac{\omega}{4}$$

$$\therefore T' = \frac{2\pi}{\omega'} = \left(\frac{8\pi}{\omega} \right)$$

$$= 4T = (4 \times 27) = 108 \text{ days}$$

Question 33: The radius of Martian orbit around the Sun is about 4 times the radius of the orbit of Mercury. The Martian year is 687 Earth days. Then which of the following is the length of 1 year on Mercury?

Options:

- (a) 172 earth days
- (b) 124 earth days
- (c) 88 earth days
- (d) 225 earth days

Answer: (c)

Solution: RM = Radius of mass

RM' = Radius of mercury

By Kepler's law: $T^2 \propto r^3$

$$RM = 4RM'$$

TM = Time period of mass.

TM' = Time period of mercury

$$\frac{TM}{TM'} = \left(\frac{RM}{RM'} \right)^{\frac{3}{2}}$$

$$= \left(\frac{4RM'}{RM'} \right)^{\frac{3}{2}} \quad TM = 687 \text{ earth days}$$

$$= 2^{2 \times \frac{3}{2}} = 2^3 = 8$$

$$\therefore \frac{TM}{TM'} = 8$$

$$\Rightarrow TM' = \frac{TM}{8} = \frac{687}{8}$$

$$\cong 88 \text{ (approximate) Earth days.}$$

Question 34: A wire of resistance R is cut into 8 equal pieces. From these pieces two equivalent resistances are made by adding four of these together in parallel. Then these two sets are added in series. The net effective resistance of the combination is:

Options:

- (a) $\frac{R}{16}$
- (b) $\frac{R}{8}$
- (c) $\frac{R}{64}$
- (d) $\frac{R}{32}$

Answer: (a)

Solution: Each wire will have resistance of $\frac{R}{8}$.

If R_{eq1} and R_{eq2} are the two sets of resistances that are made by connecting 4 in parallel.

$$\begin{aligned}\therefore \frac{1}{R_{eq1}} &= \frac{1}{R/8} + \frac{1}{R/8} + \frac{1}{R/8} + \frac{1}{R/8} \\ &= \frac{8}{R} + \frac{8}{R} + \frac{8}{R} + \frac{8}{R} = \frac{32}{R} \\ \therefore R_{eq1} &= \frac{R}{32}\end{aligned}$$

Similarly, $R_{eq2} = \frac{R}{32}$

Now, these two sets are connected in series

$$R_{eq} = R_{eq1} + R_{eq2} = \frac{R}{32} + \frac{R}{32} = \frac{R}{16}$$

Question 35: A photon and an electron (mass m) have the same energy E . The ratio ($\lambda_{\text{photon}} / \lambda_{\text{electron}}$) of their de Broglie wavelengths is: (c is the speed of light)

Options:

- (a) $c\sqrt{\frac{2m}{E}}$
- (b) $\frac{1}{c}\sqrt{\frac{E}{2m}}$
- (c) $\sqrt{\frac{E}{2m}}$
- (d) $c\sqrt{2mE}$

Answer: (a)

Solution: For electron: $\lambda_e = \frac{h}{\sqrt{2mE}}$

For photon: - $E = pc$ [$\because m_0 = \text{rest mass} = 0$]

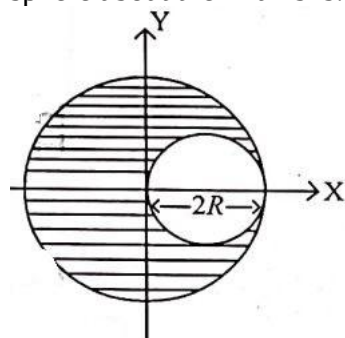
$$\lambda_{ph} = \frac{hc}{E} \left[\because E = \frac{hc}{\lambda} \right]$$

$$\therefore \frac{\lambda_{ph}}{\lambda_e} = \frac{hc}{E} \div \frac{h}{\sqrt{2mE}}$$

$$= \frac{hc}{E} \times \frac{\sqrt{2mE}}{h}$$

$$= c\sqrt{\frac{2m}{E}}$$

Question 36: A sphere of radius R is cut from a larger solid sphere of radius $2R$ as shown in the figure. The ratio of the moment of inertia of the smaller sphere to that of the rest part of the sphere about the Y-axis is:



Options:

- (a) $\frac{7}{57}$
 (b) $\frac{7}{64}$
 (c) $\frac{7}{8}$
 (d) $\frac{7}{40}$

Answer: (a)

Solution: $I''' = M.O.I$ of rest part of sphere about y-axis

$$\therefore I'' = I - I'''$$

$$= \frac{8}{5} MR^2 - \frac{7}{40} MR^2$$

$$= \frac{(64-7)MR^2}{40} = \frac{57}{40} MR^2$$

$$\therefore I'' : I''' = \frac{7}{40} \div \frac{57}{40}$$

$$= \frac{7}{40} \times \frac{40}{57} = \frac{7}{57}$$

Question 37: An electron (mass $9 \times 10^{-31} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$) moving with speed $c/100$ (c = speed of light) is injected into a magnetic field $\vec{B}$ of magnitude $9 \times 10^{-4} \text{ T}$ perpendicular to its direction of motion. We wish to apply an uniform electric $\vec{E}$ together with the magnetic field so that the electron does not deflect from its path. Then (Speed of light $c = 3 \times 10^8 \text{ ms}^{-1}$)

Options:

- (a) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is $27 \times 10^2 \text{ Vm}^{-1}$
 (b) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is $27 \times 10^4 \text{ Vm}^{-1}$
 (c) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is $27 \times 10^4 \text{ Vm}^{-1}$
 (d) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is $27 \times 10^2 \text{ Vm}^{-1}$

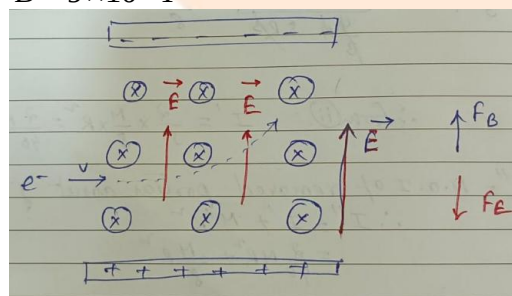
Answer: (d)

Solution:

V = velocity e^-

$$v = \frac{c}{100}$$

$$B = 9 \times 10^{-4} \text{ T}$$



F_B = Force due to magnetic field

F_E = Force due to $\vec{E}$

$$F_B = F_E$$

$$qvB = qE$$

$$\Rightarrow E = vB$$

$$\begin{aligned}
 &= \frac{c}{100} \times 9 \times 10^{-4} \\
 &= \frac{3 \times 10^8}{10^2} \times 9 \times 10^{-4} \\
 &= 27 \times 10^6 \times 10^{-4} \\
 &= 27 \times 10^2
 \end{aligned}$$

→ direction should be perpendicular to $\vec{B}$

Question 38: The electric field in a plane electromagnetic wave is given by

$$E_z = 60 \cos(5x + 1.5 \times 10^9 t) \text{ V/m}$$

Then expression for the corresponding magnetic field is (here subscripts denote the direction of the field):

Options:

- (a) $B_z = 60 \cos(5x + 1.5 \times 10^9 t) T$
- (b) $B_z = 60 \sin(5x + 1.5 \times 10^9 t) T$
- (c) $B_y = 2 \times 10^{-7} \cos(5x + 1.5 \times 10^9 t) T$
- (d) $B_x = 2 \times 10^{-7} \cos(5x + 1.5 \times 10^9 t) T$

Answer: (c)

Solution: $E_z = 60 \cos(5x + 1.5 \times 10^9 t) \text{ V/m}$

Here, $E_0 = 60$

$$B_0 = \frac{E_0}{c} = \frac{60}{3 \times 10^8} = 20 \times 10^{-8} = 2 \times 10^{-7}$$

$\vec{B}$ Should be perpendicular to $\vec{E}$

$$\therefore B_y = 2 \times 10^{-7} \cos(5x + 1.5 \times 10^9 t) T$$

Question 39: A body weighs 48 N on the surface of the earth. The gravitational force experienced by the body due to the earth at a height equal to one-third the radius of the earth from its surface is:

Options:

- (a) 32 N
- (b) 36 N
- (c) 16 N
- (d) 27 N

Answer: (d)

Solution: W = Weight on surface of earth

$$W = mg = 48 \text{ N}$$

g_h = acceleration due to gravity at height h .

$$\begin{aligned}
 \therefore g_h &= \frac{g}{\left(1 + \frac{h}{R}\right)^2} = \frac{g}{\left(1 + \frac{R}{3R}\right)^2} \\
 &= \frac{g}{\left(1 + \frac{1}{3}\right)^2} = \frac{g}{\left(\frac{4}{3}\right)^2} \\
 &= \frac{9g}{16} \\
 \therefore W' &= \text{New } W
 \end{aligned}$$

$$= mgh = \frac{m \times 9g}{16}$$

$$= \frac{9}{16} mg = \frac{9}{16} \times 48 = 27 N$$

Question 40: An unpolarized light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then -

Options:

- (a) Both reflected and transmitted light are perfectly polarized with angles of reflection and refraction close to 60° and 30° , respectively
- (b) Transmitted light is completely polarized with angle of refraction close to 30°
- (c) Reflected light is completely polarized and the angle of reflection is close to 60°
- (d) Reflected light is partially polarized and the angle of reflection is close to 30°

Answer: (c)

Solution: $\mu = 1.73$

$$\mu = \tan \theta_B$$

$$\therefore \theta_B = \tan^{-1}(1.73)$$

$$= \tan^{-1}(\sqrt{3})$$

$$= \tan^{-1}(\tan 60^\circ)$$

$$\theta_B = 60^\circ$$

Reflected light ray should be completely polarised.

θ_B = angle of incident

There, θ_B = angle of incidence

$$\therefore \text{angle of reflection} = \theta_B = 60^\circ$$

$$\therefore 1 \times \sin \theta_B = \mu \sin r$$

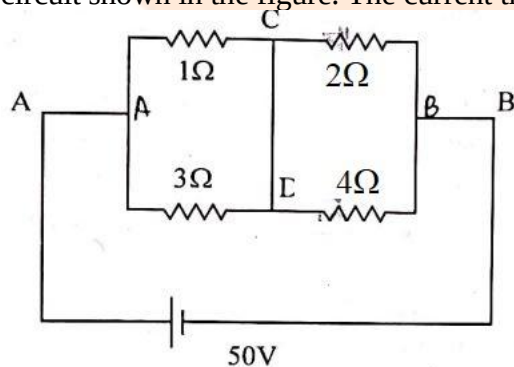
$$\Rightarrow \sin(60^\circ) = \sqrt{3} \times \sin r$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \sqrt{3} \sin r$$

$$\therefore r = 30^\circ \text{ (angle of refraction).}$$

→ Transmitted light is partially polarised.

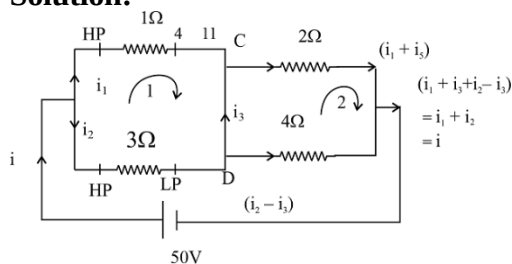
Question 41: A constant voltage of 50 V is maintained between the points A and B of the circuit shown in the figure. The current through the branch CD of the circuit is:



Options:

- (a) 2.5 A
- (b) 3.0 A
- (c) 1.5 A
- (d) 2.0 A

Answer: (d)

Solution:

$$R_{eq1} = \frac{1 \times 3}{1 + 3} = \frac{3}{4}$$

$$R_{eq2} = \frac{2 \times 4}{2 + 4} = \frac{8}{6} = \frac{4}{3}$$

$$\therefore R_{eq} = \frac{3}{4} + \frac{4}{3}$$

$$= \frac{9 + 16}{12} = \frac{25}{12} \Omega$$

$$\therefore i = \frac{50 \times 12}{25} = 24A$$

$$\text{Loop 1: } +3i_2 - 1i_1 = 0$$

$$3i_2 = i_1$$

$$i_1 + i_2 = i$$

$$\Rightarrow 3i_2 + i_2 = i$$

$$\Rightarrow 4i_2 = 24$$

$$\Rightarrow i_2 = 6A$$

$$\therefore i_1 = 3i_2 = 18A$$

$$\text{Loop 2: } 4(i_2 - i_3) - 2(i_1 + i_3) = 0$$

$$\Rightarrow 4i_2 - 4i_3 - 2i_1 - 2i_3 = 0$$

$$\Rightarrow 4i_2 - 2i_1 - 6i_3 = 0$$

$$\Rightarrow 24 - (2 \times 18) - 6i_3 = 0$$

$$\Rightarrow 6i_3 = 24 - 36$$

$$6i_3 = -12$$

$$\therefore i_3 = -2A$$

$$\therefore i_3 = -2A$$

$\therefore$ direction from C to D.

Question 42: The plates of a parallel plate capacitor are separated by d . Two slabs of different dielectric constant K_1 and K_2 with thickness $\frac{3}{8}d$ and $\frac{d}{2}$, respectively are inserted in the capacitor. Due to this, the capacitance becomes two times larger than when there is nothing between the plates.

If $K_1 = 1.25K_2$, the value of K_1 is:

Options:

(a) 1.60

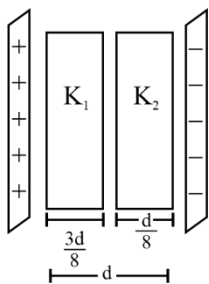
(b) 1.33

(c) 2.66

(d) 2.33

Answer: (c)

Solution:



Given $c' = 2c$

$$c = \frac{\epsilon_0 A}{d}$$

$$k_2 = \frac{K_1}{1.25} = \frac{4K_1}{5}$$

$$c' = \frac{\epsilon_0 A}{\left(d - t_1 - t_2 + \frac{t_1}{k_1} + \frac{t_2}{k_2}\right)}$$

$$c' = \frac{\epsilon_0 A}{d - \frac{3d}{8} - \frac{d}{2} + \frac{3d}{8k_1} + \frac{d}{2k_2}}$$

$$2c = \frac{\epsilon_0 A}{d \left[1 - \frac{3}{8} - \frac{1}{2} + \frac{3}{8k_1} + \frac{1}{2k_2}\right]}$$

$$2c = \frac{c}{\left[\frac{1}{8} + \frac{3}{8k_1} + \frac{5}{2 \times 4k_1}\right]}$$

$$\Rightarrow 2 = \frac{1}{\left(\frac{1}{8} + \frac{3}{8k_1} + \frac{5}{8k_1}\right)}$$

$$\Rightarrow 2 \left[\frac{k_1 + 3 + 5}{8k_1} \right] = 1$$

$$\Rightarrow 2k_1 + 16 = 8k_1$$

$$\Rightarrow 6k_1 = 16$$

$$\Rightarrow k_1 = \frac{8}{3} = 2.66$$

Question 43: Consider the diameter of a spherical object being measured with the help of a Vernier caliper. Suppose its 10 Vernier scale Divisions (V.S.D.) are equal to its 9 Main scale Divisions (M.S.D). The least division in the M.S. is 0.1 cm and the zero of V.S. is at $x = 0.1\text{cm}$ when the jaws of Vernier calipers are closed.

If the main scale reading for the diameter is $M = 5\text{ cm}$ and the number of coinciding vernier division is 8, the measured diameter after zero error correction, is

Options:

- (a) 4.98 cm
- (b) 5.00 cm
- (c) 5.18 cm
- (d) 5.08 cm

Answer: (a)

Solution: Total reading = $MSR + (VSR \times LC) - \text{zero error}$

$$= 5 + (8 \times LC) - 0.1$$

$$L.C \ 10 \ VSD = 9 \ MSD$$

$$L.C = 1MSD - 1VSD$$

$$= 1MSD - \frac{9}{10} MSD$$

$$= \frac{1}{10} MSD$$

$$= \frac{1}{10} \times 0.1cm = 0.01cm$$

$$T \text{ reading} = 5 + (8 \times 0.01) - 0.1 = 4.98$$

Question 44: A 2 amp current is flowing through two different small circular copper coils having radii ratio 1: 2. The ratio of their respective magnetic moments will be

Options:

(a) 2 : 1

(b) 4 : 1

(c) 1 : 4

(d) 1 : 2

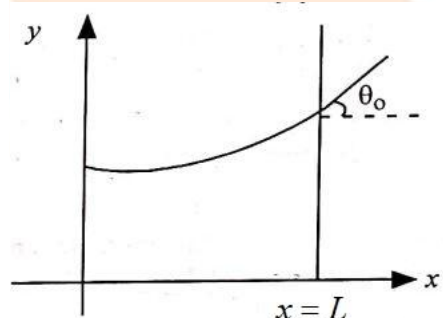
Answer: (c)

Solution: $R_1 : R_2 = 1 : 2$

$$M_1 : M_2 = iA_1 : iA_2$$

$$= R_1^2 : R_2^2 = 1 : 4$$

Question 45: Consider a water tank shown in the figure. It has one wall at $x = L$ and can be taken to be very wide in the z direction. When filled with a liquid of surface tension S and density ρ , the liquid surface makes angle θ_0 ($\theta_0 \ll 1$) with the x -axis at $x = L$. If $y(x)$ is the height of the surface then the equation for $y(x)$ is:



(take $\theta(x) = \sin \theta(x) = \tan \theta(x) = \frac{dy}{dx}$, g is the acceleration due to gravity)

Options:

(a) $\frac{d^2y}{dx^2} = \sqrt{\frac{\rho g}{S}}$

(b) $\frac{dy}{dx} = \sqrt{\frac{\rho g}{S}} x$

(c) $\frac{d^2y}{dx^2} = \frac{\rho g}{S} x$

(d) $\frac{d^2y}{dx^2} = \frac{\rho g}{S} y$

Answer: (d)

Solution: $R = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\left|\frac{d^2y}{dx^2}\right|}$

θ is very small, $\frac{dy}{dx}$ is small

$$R \approx \frac{1}{\left(\frac{d^2y}{dx^2}\right)}$$

Laplace equation : $\Delta p = \frac{T}{R}$

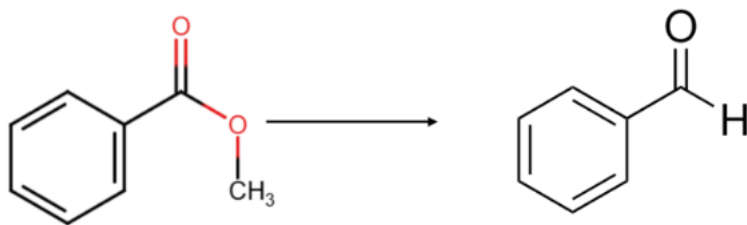
$$\Delta p = T \frac{d^2y}{dx^2}$$

$$\frac{\rho gh}{T} = \frac{d^2y}{dx^2}$$

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Chemistry

Question 46: Identify the suitable reagent for the following conversion.

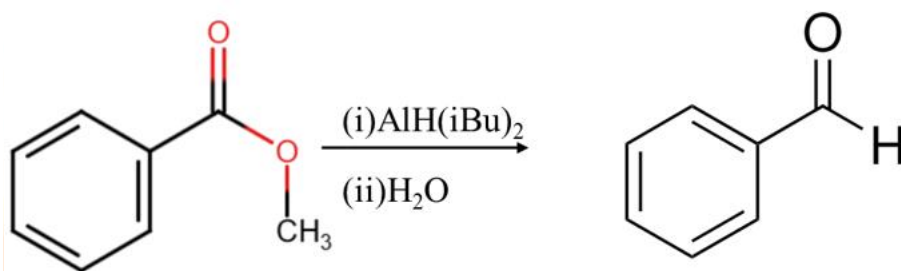


Options:

- (a) (i) NaBH_4 , (ii) $\text{H}^+/\text{H}_2\text{O}$
- (b) $\text{H}_2/\text{Pd}-\text{BaSO}_4$
- (c) (i) LiAlH_4 , (ii) $\text{H}^+/\text{H}_2\text{O}$
- (d) (i) $\text{AlH}(\text{iBu})_2$ (ii) H_2O

Answer: (d)

Solution:



Question 47: The correct order of decreasing acidity of the following aliphatic acids is

Options:

- (a) $\text{HCOOH} > \text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH}$
- (b) $\text{HCOOH} > (\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH}$
- (c) $(\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{CH}_3\text{COOH} > \text{HCOOH}$
- (d) $\text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH} > \text{HCOOH}$

Answer: (a)

Solution: $\text{HCOOH} > \text{CH}_3\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH}$
 +I Effect increases, Acidic strength decreases

Question 48: Which one of the following reactions does NOT belong to “Lassaigne’s test”?

Options:

- (a) $\text{Na} + \text{X} \xrightarrow{\Delta} \text{NaX}$
- (b) $2\text{CuO} + \text{C} \xrightarrow{\Delta} 2\text{Cu} + \text{CO}_2$
- (c) $\text{Na} + \text{C} + \text{N} \xrightarrow{\Delta} \text{NaCN}$
- (d) $2\text{Na} + \text{S} \xrightarrow{\Delta} \text{Na}_2\text{S}$

Answer: (b)

Solution: $2\text{CuO} + \text{C} \xrightarrow{\Delta} 2\text{Cu} + \text{CO}_2$

Question 49: If the rate constant of a reaction is 0.03 s^{-1} , how much time does not take for 7.2 mol L^{-1} concentration of the reactant to get reduced to 0.9 mol L^{-1}
 (Given: $\log 2 = 0.301$)

Options:

- (a) 210 s
- (b) 21.0 s
- (c) 69.3 s
- (d) 23.1 s

Answer: (c)

Solution:

Rate constant $k = 0.03 \text{ s}^{-1}$

Initial concentration $[A]_0 = 7.2 \text{ mol L}^{-1}$

Final concentration $[A] = 0.9 \text{ mol L}^{-1}$

$\log 2 = 0.301$

For a first-order reaction, the integrated rate law is:

$$\ln [A] - \ln [A]_0 = -kt$$

$$t = \frac{1}{k} \times 2.303 \log \frac{[A]_0}{[A]}$$

$$t = \frac{1}{0.03} \times 2.303 \log \frac{7.2}{0.9}$$

$$t = \frac{1}{0.03} \times 2.303 \log 8$$

$$t = \frac{1}{0.03} \times 2.303 \times \log 2$$

$$t = \frac{1}{0.03} \times 2.303 \times 0.3010 = 69.3 \text{ s}$$

Question 50: Given below are two statements :

Statement I: A hypothetical diatomic molecule with bond order zero is quite stable.

Statement II: As bond order increases, the bond length increases.

In the light of the above statements, choose the most appropriate answer from the options given below :

Options:

- (a) Statement I is true but Statement II is false
- (b) Statement I is false but Statement II is true
- (c) Both Statement I and Statement II are true
- (d) Both Statement I and Statement II are false

Answer: (d)

Solution: Both Statement I and Statement II are false

Diatomic molecule cannot have bond order zero and it's unstable

Bond order increases, the bond length decreases.

Question 51: Out of the following complex compounds, which of the compound will be having the minimum conductance in solution?

Options:

- (a) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- (b) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}$
- (c) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
- (d) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$

Answer: (d)

Solution: $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ is not producing any ions in solution so it has less conductance.

Question 52: Which of the following aqueous solution will exhibit highest boiling point?

Options:

- (a) 0.01 M Na_2SO_4
- (b) 0.015 M $\text{C}_6\text{H}_{12}\text{O}_6$
- (c) 0.01 M Urea

(d) 0.01 M KNO_3

Answer: (a)

Solution:

(a) 0.01 M $\text{Na}_2\text{SO}_4 = i = 3$

(b) 0.015 M $\text{C}_6\text{H}_{12}\text{O}_6 = i = 1$

(c) 0.01 M Urea = $i = 1$

(d) 0.01 M $\text{KNO}_3 = i = 2$

$$\Delta T_b = i k_b m$$

$$T_b = i k_b m + T_b^\circ$$

More the value of i more will be boiling point.

Question 53: Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A):  I undergoes $\text{S}_\text{N}2$ reaction faster than  Cl

Reason (R): Iodine is a better leaving group because of its large size.

In the light of the above statements, choose the correct answer from the options given below.

Options:

(a) A is true but R is false

(b) A is false but R is true

(c) Both A and R are true and R is the correct explanation of A

(d) Both A and R are true but R is not the correct explanation of A

Answer: (c)

Solution: Both A and R are true and R is the correct explanation of A

Since, both the given halides are primary and iodine is a better leaving group because of charge dispersion

Question 54: Consider the following compounds :

KO_2 , H_2O_2 and H_2SO_4 .

The oxidation states of the underlined elements in them are, respectively,

Options:

(a) +1, -2, and +4

(b) +4, -4, and +6

(c) +1, -1, and +6

(d) +2, -2, and +6

Answer: (c)

Solution:

KO_2 , it is super oxide, Oxidation number of K = +1

H_2O_2 , peroxide oxidation number -1

H_2SO_4 , $= 2 \times (+1) + \text{S} + 4 \times (-2) = 0$

S = +6

Question 55: Match List-I with List-II.

List-I	List-II
A. Haber process	I. Fe catalyst
B. Wacker oxidation	II. PdCl_2
C. Wilkinson catalyst	III. $[(\text{PPh}_3)_3\text{RhCl}]$
D. Ziegler catalyst	IV. TiCl_4 with $\text{Al}(\text{CH}_3)_3$

Choose the correct answer from the options given below:

Options:

(a) A-I, B-II, C-III, D-IV

(b) A-I, B-IV, C-III, D-II

(c) A-I, B-II, C-IV, D-III

(d) A-II, B-III, C-I, D-IV

Answer: (a)

Solution: A-I, B-II, C-III, D-IV

Question 56: Given below are two statements :

Statement I: Like nitrogen that can form ammonia, arsenic can form arsine.

Statement II: Antimony cannot form antimony pentoxide.

In the light of the above statements, choose the most appropriate answer from the options given below :

Options:

- (a) Statement I is correct but Statement II is incorrect
- (b) Statement I is incorrect but Statement II is correct
- (c) Both Statement I and Statement II are correct
- (d) Both Statement I and Statement II are incorrect

Answer: (a)

Solution: Statement I is correct but Statement II is incorrect.

Arsenic can form arsine

Antimony can form antimony pentoxide.

Question 57: Given below are two statements :

Statement I: Ferromagnetism is considered as an extreme form: of paramagnetism.

Statement II: The number of unpaired electrons in a Cr^{2+} ion ($Z = 24$) is the same as that of a Nd^{3+} for ($Z = 60$).

In the light of the above statements, choose the correct answer from the options given below:

Options:

- (a) Statement I is true but Statement II is false
- (b) Statement I is false but Statement II is true
- (c) Both Statement I and Statement II are true
- (d) Both Statement I and Statement II are false

Answer: (a)

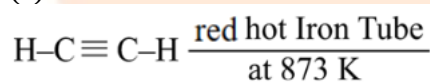
Solution: Statement I is true but Statement II is false

Cr^{2+} is $3d^4$ having 4 unpaired electrons and Nd^{3+} is $4f^3$ having 3 unpaired electrons

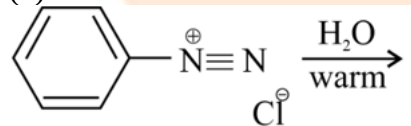
Question 58: Which one of the following reactions does Not give benzene as the product?

Options:

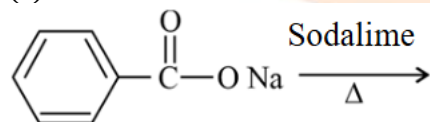
(a)



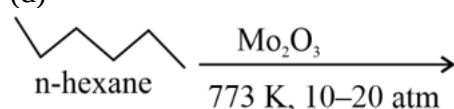
(b)



(c)

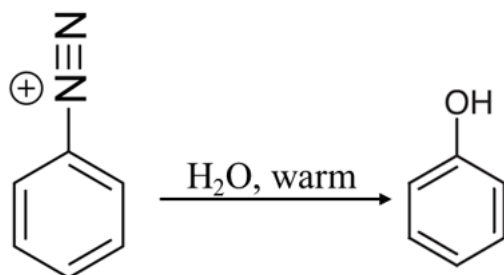


(d)

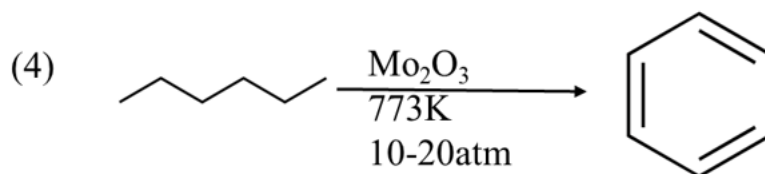
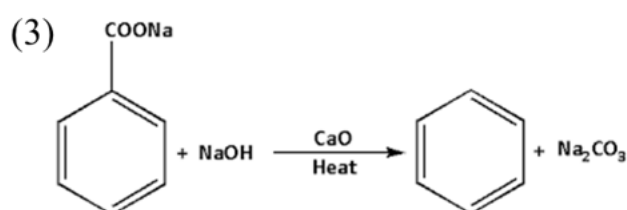
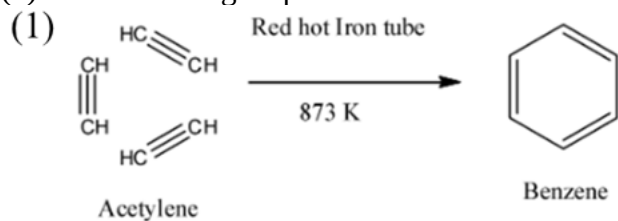


Answer: (b)

Solution:



(2) Reaction will give phenol rather than Benzene



Question 59: Match List-I with List-II

List-I	List-II
A. XeO_3	I. sp^3d ; linear
B. XeF_2	II. sp^3 ; pyramidal
C. XeOF_4	III. sp^3d^3 ; distorted octahedral
D. XeF_6	IV. sp^3d^2 ; square pyramidal

Choose the correct answer from the options given below:

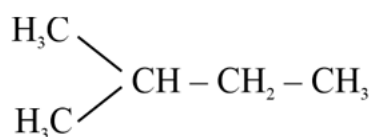
Options:

- (a) A-IV, B-II, C-III, D-I
- (b) A-IV, B-II, C-I, D-III
- (c) A-II, B-I, C-IV, D-III
- (d) A-II, B-I, C-III, D-IV

Answer: (c)

Solution: A-II, B-I, C-IV, D-III

Question 60: How many products (including stereoisomers) are expected from monochlorination of the following compound?

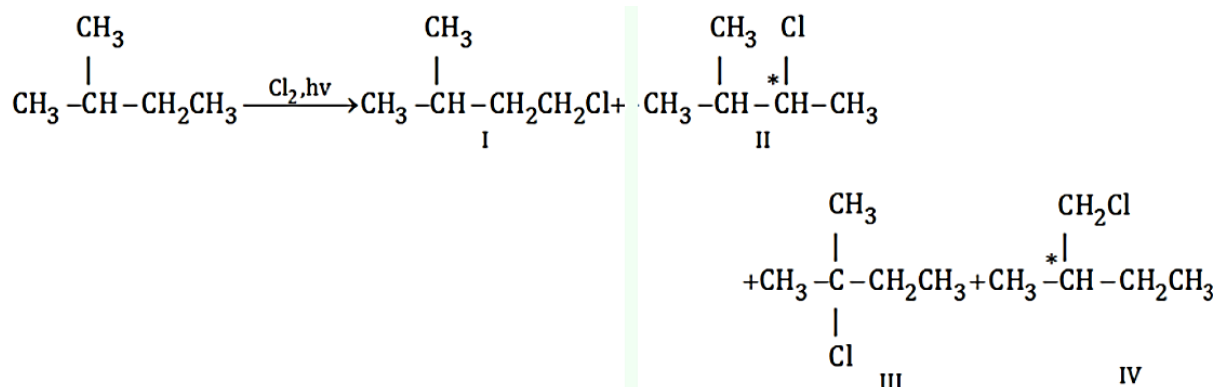


Options:

- (a) 5
- (b) 6
- (c) 2
- (d) 3

Answer: (b)

Solution:



Question 61: Which of the following statements are true?

- A. Unlike Ga that has a very high melting point, Cs has a very low melting point
- B. On Pauling scale, the electronegativity values of N and Cl are not the same.
- C. Ar, K^+ , Cl^- , Ca^{2+} , and S^{2-} are all isoelectronic species.
- D. The correct order of the first ionization enthalpies of Na, Mg, Al, and Si is $\text{Si} > \text{Al} > \text{Mg} > \text{Na}$.
- E. The atomic radius of Cs is greater than that of Li and Rb.

Choose the correct answer from the options given below

Options:

- (a) C and D only
- (b) A, C, and E only
- (c) A, B, and E only
- (d) C and E only

Answer: (d)

Solution:

- A) Ga melting point is not high.
 - B) N has E.N = 3.04, Cl has E.N = 3.16
 - C) Ar, K^+ , Cl^- , Ca^{2+} , and S^{2-} are all isoelectronic species having 18 electrons
 - D) $\text{Si} > \text{Mg} > \text{Al} > \text{Na}$
 - E) The atomic radius of Cs is greater than that of Li and Rb.
- Size increases down the group

Question 62: The standard heat of formation, in kcal/mol of Ba^{2+} is :

[Given: standard heat of formation of SO_4^{2-} ion (aq) = -216 kcal/mol, standard heat of crystallisation of $\text{BaSO}_4(\text{s}) = -4.5$ kcal/mol, standard heat of formation of $\text{BaSO}_4(\text{s}) = -349$ kcal/mol]

Options:

- (a) $+133.0$
- (b) $+220.5$
- (c) -128.5
- (d) -133.0

Answer: (c)

Solution:

$$\Delta H_f^\circ(\text{Ba}^{2+}) = \Delta H_f^\circ(\text{BaSO}_4) - \Delta H_{\text{cryst}}^\circ(\text{BaSO}_4) - \Delta H_f^\circ(\text{SO}_4^{2-})$$

$$\Delta H_f^\circ(\text{Ba}^{2+}) = -349 \text{ kcal/mol} - (-4.5 \text{ kcal/mol}) - (-216 \text{ kcal/mol})$$

$$\Delta H_f^\circ(\text{Ba}^{2+}) = -349 \text{ kcal/mol} + 4.5 \text{ kcal/mol} + 216 \text{ kcal/mol}$$

$$\Delta H_f^\circ(\text{Ba}^{2+}) = -128.5 \text{ kcal/mol}$$

Question 63: Match List-I with List-II

List-I (Example)

List-II (Type of Solution)

A. Humidity	I. Solid in solid
B. Alloys	II. Liquid in gas
C. Amalgams	III. Solid in gas
D. Smoke	IV. Liquid in solid

Choose the correct answer from the options given below :

Options:

(a) A-III, B-I, C-IV, D-II

(b) A-III, B-II, C-I, D-IV

(c) A-II, B-IV, C-I, D-III

(d) A-II, B-I, C-IV, D-III

Answer: (d)

Solution: A-II, B-I, C-IV, D-III

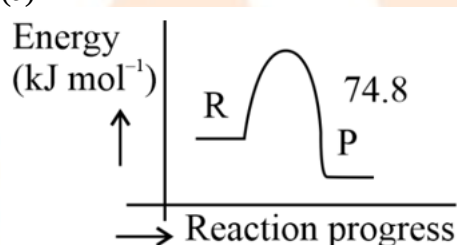
Question 64: $C(s) + 2H_2(g) \rightarrow CH_4(g)$; $\Delta H = -74.8 \text{ kJ mol}^{-1}$

Which of the following diagrams gives an accurate representation of the above reaction?

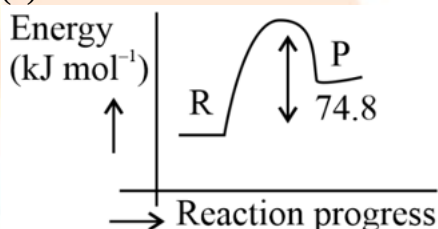
[R $\rightarrow$ reactants; P $\rightarrow$ products]

Options:

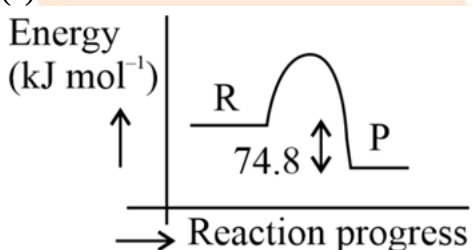
(a)



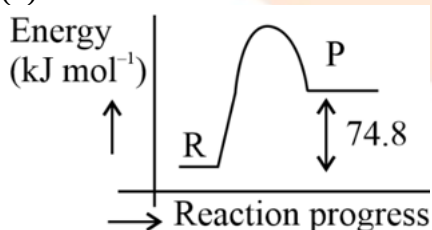
(b)



(c)

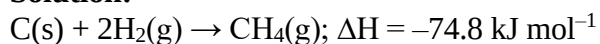


(d)

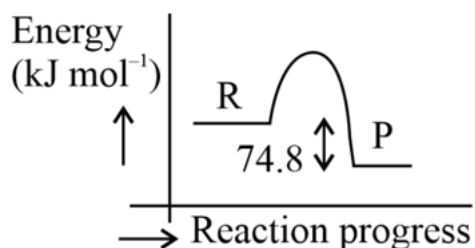


Answer: (c)

Solution:



It is an exothermic process. So energy is released therefore for products energy will be less

**Question 65:**

Sugar 'X'

- A. is found in honey
- B. is a keto sugar.
- C. exists in α and β – anomeric forms
- D. is laevorotatory

'X' is

Options:

- (a) Maltose
- (b) Sucrose
- (c) D-Glucose
- (d) D-Fructose

Answer: (d)

Solution: Honey contains Glucose and fructose. Fructose is a ketohexose. Fructose exist in α -D-fructofuranose and β -D-fructofuranose which are anomers.

Fructose is levorotatory

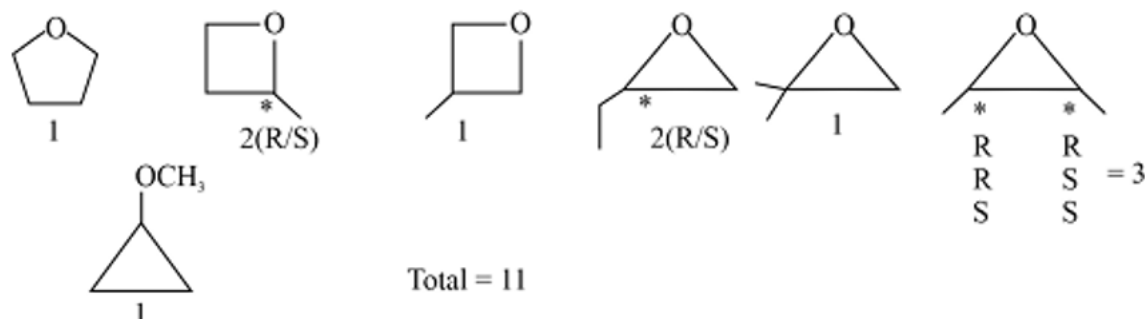
Question 66: Total number of possible isomers (both structural as well as stereoisomers) of cyclic ethers of molecular formula C_4H_8O is :

Options:

- (a) 10
- (b) 11
- (c) 6
- (d) 8

Answer: (b)**Solution:**

C_4H_8O - Cyclic ether isomers

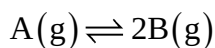


Question 67: For the reaction $A(g) \rightleftharpoons 2B(g)$, the backward reaction rate constant is higher than the forward reaction rate constant by a factor of 2500, at 1000 K.

[Given: $R = 0.0831 \text{ L atm mol}^{-1} \text{ K}^{-1}$] K_p for the reaction at 1000 K is

Options:

- (a) 0.033
- (b) 0.021
- (c) 83.1
- (d) 2.077×10^5

Answer: (a)**Solution:**

$$r_f = K_f [A]$$

$$r_b = K_b [B]^2$$

At eq.

$$r_f = r_b$$

$$K_f [A] = K_b [B]^2$$

$$K_f [A] = 2500 \times K_b [B]^2$$

$$\frac{K_f}{2500 \times K_b} = \frac{[B]^2}{[A]}$$

$$K_c = \frac{1}{2500} \text{ mol / lit}$$

$$K_p = K_c [RT]^{\Delta n_g}$$

$$K_p = 0.03324 \text{ atm}$$

Question 68: The ratio of the wavelengths of the light absorbed by a Hydrogen atom when it undergoes $n = 2 \rightarrow n = 3$ and $n = 4 \rightarrow n = 6$ transitions, respectively, is

Options:

(a) $\frac{1}{9}$

(b) $\frac{1}{4}$

(c) $\frac{1}{36}$

(d) $\frac{1}{16}$

Answer: (b)**Solution:**

$$\frac{1}{\lambda_1} = R \times Z^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \dots (1)$$

$$\frac{1}{\lambda_2} = R \times Z^2 \left(\frac{1}{4^2} - \frac{1}{6^2} \right) \dots (2)$$

$$\frac{\lambda_1}{\lambda_2} = \frac{\left(\frac{1}{4^2} - \frac{1}{6^2} \right)}{\left(\frac{1}{2^2} - \frac{1}{3^2} \right)} = \frac{\frac{1}{16} - \frac{1}{36}}{\frac{1}{4} - \frac{1}{9}} = \frac{1}{4}$$

Question 69: If the molar conductivity (Λ_m) of a 0.050 mol L^{-1} solution of a monobasic weak acid is $90 \text{ S cm}^2 \text{ mol}^{-1}$, its extent (degree) of dissociation will be
[Assume $\Lambda^{o+} = 349.6 \text{ S cm}^2 \text{ mol}^{-1}$ and $\Lambda^{o-} = 50.4 \text{ S cm}^2 \text{ mol}^{-1}$]

Options:

(a) 0.225

(b) 0.215

(c) 0.115

(d) 0.125

Answer: (a)

Solution:

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0}$$

$$\Lambda_m^0 = 349.6 + 50.4 = 400 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{90}{400} = 0.225$$

Question 70: 5 moles of liquid X and 10 moles of liquid Y make a solution having a vapour pressure of 70 torr. The vapour pressures of pure X and Y are 63 torr and 78 torr respectively. Which of the following is true regarding the described solution?

Options:

- (a) The solution is ideal.
- (b) The solution has volume greater than the sum of individual volumes.
- (c) The solution shows positive deviation.
- (d) The solution shows negative deviation.

Answer: (d)

Solution: Liquid X = 5 moles

Liquid Y = 10 moles

$P_T = 70$ torr

$$P_T = P_x^0 X_x + P_y^0 Y_y$$

$$= 63 \times \frac{1}{3} + 78 \times \frac{2}{3} = 73$$

As V.P is less than 73 that is 70 negative deviation

Question 71: Among the following, choose the ones with equal number of atoms.

- A. 212 g of Na_2CO_3 (s) [molar mass = 106 g]
- B. 248 g of Na_2O (s) [molar mass = 62 g].
- C. 240 g of NaOH (s) [molar mass = 40 g]
- D. 12 g of H_2 (g) [molar mass = 2 g]
- E. 220 g of CO_2 (g) [molar mass = 44 g]

Choose the correct answer from the options given below :

Options:

- (a) B, C, and D only
- (b) B, D, and E only
- (c) A, B, and C only
- (d) A, B, and D only

Answer: (d)

Solution:

- A. 212 g of Na_2CO_3 (s) [molar mass = 106 g] = 2 moles = $2 \times 6 N_A$ atoms
- B. 248 g of Na_2O (s) [molar mass = 62 g] = 4 moles = $4 \times 3 N_A$ atoms
- C. 240 g of NaOH (s) [molar mass = 40 g] = 6 moles = $6 \times 3 N_A$ atoms
- D. 12 g of H_2 (g) [molar mass = 2 g] = 6 moles = $6 \times 2 N_A$ atoms
- E. 220 g of CO_2 (g) [molar mass = 44 g] = 5 moles = $5 \times 3 N_A$ atoms

Question 72: Which of the following are paramagnetic?

- A. $[\text{NiCl}_4]^{2-}$
- B. $\text{Ni}(\text{CO})_4$
- C. $[\text{Ni}(\text{CN})_4]^{2-}$
- D. $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
- E. $\text{Ni}(\text{PPh}_3)_4$

Choose the correct answer from the options given below:

Options:

- (a) A and D only
- (b) A, D and E only

(c) A and C only

(d) B and E only

Answer: (a)

Solution: A. $[\text{NiCl}_4]^{2-} = \text{Ni}^{2+}$ is $3d^8$ paramagnetic

D. $[\text{Ni}(\text{H}_2\text{O})_6]^{2+} = \text{Ni}^{2+}$ is $3d^8$ paramagnetic

Question 73: If the half-life ($t_{1/2}$) for a first order reaction is 1 minute, then the time required for 99.9% completion of the reaction is closest to

Options:

(a) 5 minutes

(b) 10 minutes

(c) 2 minutes

(d) 4 minutes

Answer: (b)

Solution:

$$t_{99.9} = 10 \times t_{1/2}$$

$$t_{99.9} = 10 \text{ min}$$

Question 74: Energy and radius of Bohr Orbit of He^+ and Li^{2+} are

[Given $R_H = 2.18 \times 10^{-18} \text{ J}$, $a_0 = 52.9 \text{ pm}$]

Options:

(a) $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-16} \text{ J}$; $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$, $E_n(\text{He}^+) = 8.72 \times 10^{-16} \text{ J}$, $r_n(\text{He}^+) = 26.4 \text{ pm}$

(b) $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-16} \text{ J}$; $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$, $E_n(\text{He}^+) = -19.62 \times 10^{-16} \text{ J}$, $r_n(\text{He}^+) = 17.6 \text{ pm}$

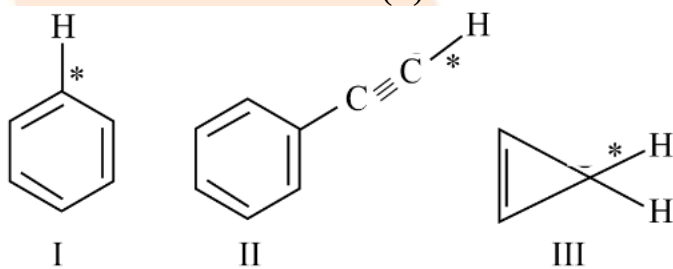
(c) $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-18} \text{ J}$; $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$, $E_n(\text{He}^+) = -8.72 \times 10^{-18} \text{ J}$, $r_n(\text{He}^+) = 26.4 \text{ pm}$

(d) $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-18} \text{ J}$; $r_n(\text{Li}^{2+}) = 26.4 \text{ pm}$, $E_n(\text{He}^+) = -19.62 \times 10^{-18} \text{ J}$, $r_n(\text{He}^+) = 17.6 \text{ pm}$

Answer: (c)

Solution: $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-18} \text{ J}$; $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$, $E_n(\text{He}^+) = -8.72 \times 10^{-18} \text{ J}$, $r_n(\text{He}^+) = 26.4 \text{ pm}$

Question 75: Among the given compounds I-III, the correct order of bond dissociation energy of C—H bond marked with * (is).



Options:

(a) $\text{III} > \text{II} > \text{I}$

(b) $\text{II} > \text{III} > \text{I}$

(c) $\text{II} > \text{I} > \text{III}$

(d) $\text{I} > \text{II} > \text{III}$

Answer: (c)

Solution: Bond dissociation energy is inversely proportional to stability of free radicals

Stability order $\text{III} > \text{I} > \text{II}$

B.D.E = $2 > 1 > 3$

Question 76: Dalton's Atomic theory could not explain which of the following?

Options:

(a) Law of multiple proportion

(b) Law of gaseous volume

(c) Law of conservation of mass

(d) Law of constant proportion

Answer: (b)

Solution: Law of gaseous volume

Question 77: Identify the correct orders against the property mentioned

- A. $\text{H}_2\text{O} > \text{NH}_3 > \text{CHCl}_3$ – dipole moment
- B. $\text{XeF}_4 > \text{XeO}_3 > \text{XeF}_2$ – number of lone pairs on central atom
- C. $\text{O-H} > \text{C-H} > \text{N-O}$ – bond length
- D. $\text{N}_2 > \text{O}_2 > \text{H}_2$ – bond enthalpy

Choose the answer from the options given below :

Options:

- (a) A, C only
- (b) B, C only
- (c) A, D only
- (d) B, D only

Answer: (c)

Solution:

$\text{H}_2\text{O} > \text{NH}_3 > \text{CHCl}_3$ – dipole moment

$\text{N}_2 > \text{O}_2 > \text{H}_2$ – bond enthalpy

Question 78: Match List I with List II.

List I (Name of Vitamin)	List II (Deficiency disease)
A. Vitamin B ₁₂	I. Cheilosis
B. Vitamin D	II. Convulsions
C. Vitamin B ₂	III. Rickets
D. Vitamin B ₆	IV. Pernicious anaemia

Choose the correct answer from the options given below:

Options:

- (a) A-II, B-III, C-I, D-IV
- (b) A-IV, B-III, C-II, D-I
- (c) A-I, B-III, C-II, D-IV
- (d) A-IV, B-III, C-I, D-II

Answer: (d)

Solution: A-IV, B-III, C-I, D-II

Question 79: The correct order of decreasing basic strength of the given amine is :

Options:

- (a) N-ethylethangmine > ethanamine > N-methylaniline > benzenamine
- (b) Benzenamine > ethanamine > N-methylaniline > N-ethylethanamine
- (c) N-methylaniline > benzenamine > ethanamine > N-ethylethanamine
- (d) N-ethylethanamine > ethanamine > benzenamine > N-methylaniline

Answer: (a)

Solution: N-ethylethangmine > ethanamine > N-methylaniline > benzenamine

Question 80: The correct order of the wavelength of light absorbed by the following complexes is,

- A. $[\text{Co}(\text{NH}_3)_6]^{3+}$
- B. $[\text{Co}(\text{CN})_6]^{3-}$
- C. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$
- D. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$

Choose the correct answer from the options given below:

Options:

- (a) $\text{C} < \text{D} < \text{A} < \text{B}$
- (b) $\text{C} < \text{A} < \text{D} < \text{B}$
- (c) $\text{B} < \text{D} < \text{A} < \text{C}$

(d) $B < A < D < C$

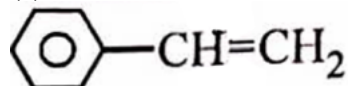
Answer: (d)

Solution: $B < A < D < C$

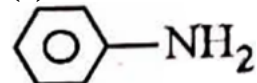
Question 81: Which one of the following compounds does not decolourize bromine water?

Options:

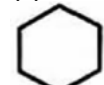
(a)



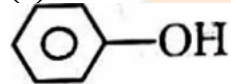
(b)



(c)

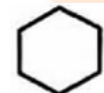


(d)

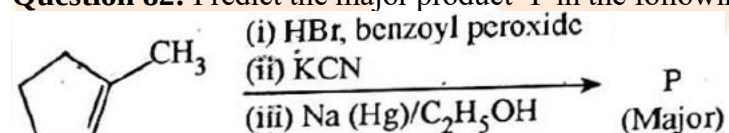


Answer: (c)

Solution:

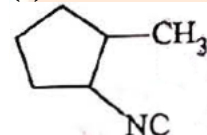


Question 82: Predict the major product 'P' in the following sequence of reactions

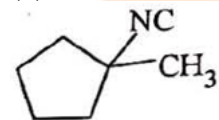


Options:

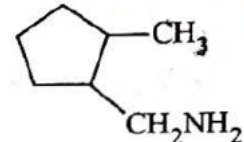
(a)



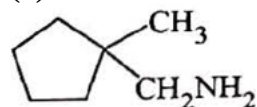
(b)



(c)

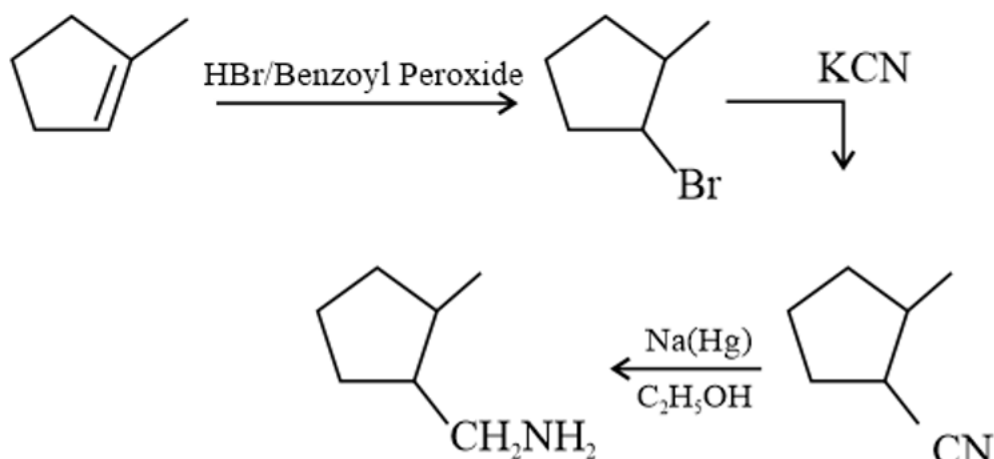


(d)



Answer: (c)

Solution:



Question 83: Match List I with List II

List I (Mixture)	List II (Method of Separation)
A. $\text{CHCl}_3 + \text{C}_6\text{H}_5\text{NH}_2$	I. Distillation under reduced pressure
B. Crude oil in petroleum industry	II. Steam distillation
C. Glycerol from spent-lye	III. Fractional distillation
D. Aniline - water	IV. Simple distillation

Choose the correct answer from the options given below :

Options:

- (a) A-III, B-IV, C-I, D-II
- (b) A-III, B-IV, C-II, D-I
- (c) A-IV, B-III, C-I, D-II
- (d) A-IV, B-III, C-II, D-I

Answer: (c)

Solution: A-IV, B-III, C-I, D-II

Question 84: Which among the following electronic configurations belong to main group elements?

- A. $[\text{Ne}]3s^1$
- B. $[\text{Ar}]3d^34s^2$
- C. $[\text{Kr}]4d^{10}5s^25p^5$
- D. $[\text{Ar}]3d^{10}4s^1$
- E. $[\text{Rn}]5f^06d^27s^2$

Choose the correct answer from the option given below

Options:

- (a) D and E only
- (b) A, C and D only
- (c) B and E only
- (d) A and C only

Answer: (c)

Solution:

- B. $[\text{Ar}]3d^34s^2$
- E. $[\text{Rn}]5f^06d^27s^2$

Question 85: Which one of the following compounds can exist as cis-trans isomers?

Options:

- (a) 1,1-Dimethylcyclopropane
- (b) 1,2-Dimethylcyclohexane
- (c) Pent-1-ene
- (d) 2-Methylhex-2-ene

Answer: (b)

Solution: 1,2-Dimethylcyclohexane

Question 86: Phosphoric acid ionizes in three steps with their ionization constant values

K_{a_1} , K_{a_2} and K_{a_3} , respectively

while K is the overall ionization constant.

Which of the following statements are true?

A. $\log K = \log K_{a_1} + K_{a_2} + K_{a_3}$

B. H_3PO_4 is a stronger acid than $H_2PO_4^-$ and HPO_4^{2-}

C. $K_{a_1} > K_{a_2} > K_{a_3}$

D. $K_{a_1} = \frac{K_{a_3} + K_{a_2}}{2}$

Choose the correct answer from the options given below:

Options:

(a) B, C and D only

(b) A, B and C only

(c) A and B only

(d) A and C only

Answer: (b)

Solution:

A. $\log K = \log K_{a_1} + K_{a_2} + K_{a_3}$

B. H_3PO_4 is a stronger acid than $H_2PO_4^-$ and HPO_4^{2-}

C. $K_{a_1} > K_{a_2} > K_{a_3}$

Question 87: Match List I with List II

List-I (Ion)	List-II (Group Number in Cation Analysis)
A. Co^{2+}	I. Group-I
B. Mg^{2+}	I. Group-III
C. Pb^{2+}	I. Group-IV
D. Al^{3+}	I. Group-VI

Choose the correct answer from the options given below.

Options:

(a) A-III, B-II, C-IV, D-I

(b) A-III, B-II, C-I, D-IV

(c) A-III, B-IV, C-II, D-I

(d) A-III, B-IV, C-I, D-II

Answer: (d)

Solution: A-III, B-IV, C-I, D-II

Question 88: Higher yield of NO in

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ can be obtained at

$[\Delta H \text{ of the reaction} = +180.7 \text{ kJ mol}^{-1}]$

A. highest temperature

B. lower temperature

C. higher concentration of N_2

D. higher concentration of O_2

Choose the correct answer from the options given below.

Options:

(a) B, C, D only

(b) A, C, D only

(c) A, D only

(d) B, C only

Answer: (b)

Solution:

According to Le-chatlier Principle

- A. highest temperature
- C. higher concentration of N_2
- D. higher concentration of O_2

Question 89: Given below are two statements:

Statement I: Benzenediazonium salt is prepared by the reaction of aniline with acid at 273 - 278 K. It decomposes easily in the dry state.

Statement II: Insertion of iodine into the benzene ring is difficult and hence iodobenzene is prepared through the reaction of benzenediazonium salt with KI.

in the light of the above statements, choose the most appropriate answer from the options given below:

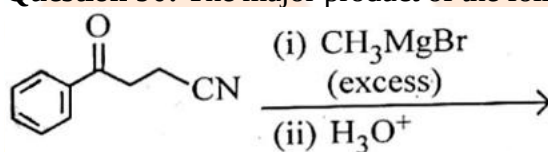
Options:

- (a) Statement I is correct but Statement II is incorrect
- (b) Statement I is incorrect but Statement II is correct
- (c) Both Statement I and Statement II are correct
- (d) Both Statement I and Statement II are incorrect

Answer: (a)

Solution: Statement I is correct but Statement II is incorrect

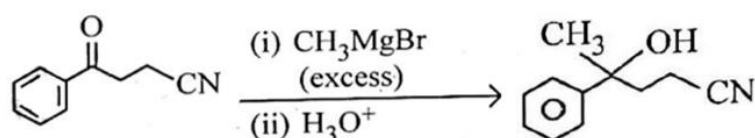
Question 90: The major product of the following reaction is:

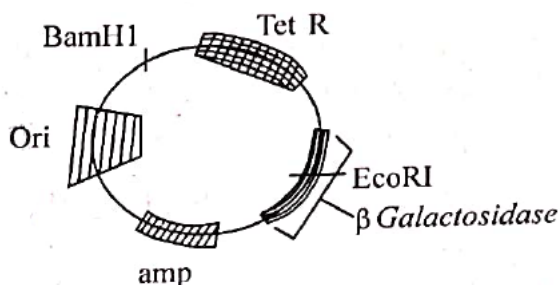
**Options:**

- (a)
- (b)
- (c)
- (d)

Answer: (d)

Solution:



NEET 2025 - Paper Code - Narmada 48**Biology****Question 91:**

In the above represented plasmid an alien piece of DNA is inserted at EcoRI site. Which of the following strategies will be chosen to select the recombinant colonies?

Options:

- (a) White color colonies will be selected
- (b) Blue color colonies grown on ampicillin
- (c) Using ampicillin & tetracyclin containing medium plate.
- (d) Blue color colonies will be selected.

Answer: (a)**Solution:**

Beta-galactosidase when activated produces a protein which reacts with chromogenic substance and gives blue colour. When DNA is inserted at EcoRI site it will cause insertional inactivation of Beta-galactosidase, thus no protein formation and reaction will take place. Resultant colonies will have white colour.

Question 92: The protein portion of an enzyme is called:**Options:**

- (a) Apoenzyme
- (b) Prosthetic group
- (c) Cofactor
- (d) Coenzyme

Answer: (a)**Solution:**

Enzymes are composed of one or several polypeptide chains. However, there are a number of cases in which non-protein constituents called co-factors are bound to the enzyme to make the enzyme catalytically active. In these instances, the protein portion of the enzymes is called the apoenzyme.

Question 93: Given below are two statements:**Statement I:** The primary source of energy in an ecosystem is solar energy.**Statement II:** The rate of production of organic matter during photosynthesis in an ecosystem is called net primary productivity (NPP).

In the light of the above statements, choose the **most appropriate** answer from the options given below:

Options:

- (a) Statement I is correct but statement II is incorrect
- (b) Statement I is incorrect but statement II is correct

- (c) Both statement I and statement II are correct
 (d) Both statement I and statement II are incorrect

Answer: (a)

Solution:

Statement 1: True. Solar energy is the ultimate source of energy for almost all ecosystems on Earth, as it drives the process of photosynthesis.

Statement 2 : False. This defines gross primary productivity (GPP). Net primary productivity (NPP) is GPP minus the energy used by producers for respiration.

Question 94: Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): A typical unfertilized, angiosperm embryo sac at maturity is 8 nucleate and 7-celled.

Reason (R): The egg apparatus has 2 polar nuclei.

In the light of the above statements, choose the correct answer from the options given below:

Options:

- (a) A is true but R is false
 (b) A is false but R is true
 (c) Both A and R are true and R is the correct explanation of A
 (d) Both A and R are true but R is not the correct explanation of A

Answer: (a)

Solution:

Assertion : True, the mature embryo sac usually have, 3 cells in egg apparatus (1 egg + 2 synergids), 3 antipodals, 1 central cell with two polar nuclei. Reason : False, egg apparatus has 3 cells (one egg and two synergids) each with one nuclei.

Question 95: Neoplastic characteristics of cells refer to:

- A. A mass of proliferating cell
 B. Rapid growth of cells
 C. Invasion and damage to the surrounding tissue
 D. Those confined to original location
 Choose the correct answer from the options given below

Options:

- (a) A, B, D only
 (b) B, C, D only
 (c) A, B only
 (d) A, B, C only

Answer: (d)

Solution:

The malignant tumors are a mass of proliferating cells called neoplastic or tumor cells. These cells grow very rapidly, invading and damaging the surrounding normal tissues. Cells sloughed from these tumors reach distant sites through blood, and wherever they get lodged in the body, they start a new tumor there (metastasis).

Question 96: Which one of the following is the characteristic feature of gymnosperms?

Options:

- (a) Seeds are absent
 (b) Gymnosperms have flowers for reproduction
 (c) Seeds are enclosed in fruits
 (d) Seeds are marked

Answer: (d)

Solution:

The gymnosperms (gymnos : naked, sperma : seeds) are plants in which the ovules are not enclosed by any ovary wall and remain exposed, both before and after fertilisation. The seeds that develop post-fertilisation, are not covered, i.e., are naked.

Question 97: Match **List-I** with **List-II**.

List-I	List-II
A. Progesterone	I. Pars intermedia
B. Relaxin	II. Ovary
C. Melanocyte stimulating hormone	III. Adrenal Medulla
D. Catecholamines	IV. Corpus luteum

Choose the correct answer from the options given below:

Options:

- (a) A-II, B-IV, C-I, D-III
- (b) A-III, B-II, C-IV, D-I
- (c) A-IV, B-II, C-I, D-II
- (d) A-IV, B-II, C-III, D-I

Answer: (c)

Solution:

Corpus luteum releases progesterone in the luteal phase of the menstrual cycle. Ovary releases relaxin during the later stages of pregnancy. Pars intermedia of Anterior Pituitary Gland releases the hormone MSH - Melanocyte Stimulating Hormone and Catecholamines (Adrenaline and Nor-adrenaline) are released by Adrenal medullary gland.

Question 98: Which chromosome in the human genome has the highest number of genes?

Options:

- (a) Chromosome 1
- (b) Chromosome 10
- (c) Chromosome X
- (d) Chromosome Y

Answer: (a)

Solution:

Chromosome 1 has most genes (2968), and the Y has the fewest (231).

Question 99: Which of the following statements about RuBisCO is true?

Options:

- (a) It is an enzyme involved in the photolysis of water.
- (b) It catalyzes the carboxylation of RuBP.
- (c) It is active only in the dark.
- (d) It has higher affinity for oxygen than carbon dioxide.

Answer: (b)

Solution:

Water Splitting Complex is involved in Photolysis. RuBisCO enzyme is involved in carboxylation of RuBP in C₃ cycle. It is involved in dark reactions but can occur in dark and light conditions both. It has higher affinity for carbon dioxide than oxygen.

Question 100: The first menstruation is called:

Options:

- (a) Diapause
- (b) Ovulation
- (c) Menopause

(d) Menarche

Answer: (d)

Solution:

The first menstruation begins at puberty and is called as menarche.

Question 101: Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

Options:

- (a) Virus
- (b) Phage
- (c) Bacterium
- (d) Yeast

Answer: (c)

Solution:

The genetically engineered organism used by Eli Lilly to prepare human insulin was *E. coli* (a bacteria).

Question 102: Given below are two statements: one is labelled as **Assertion (A)** and the other labelled as **Reason (R)**.

Assertion (A): All vertebrates are chordates but all chordates are not vertebrate.

Reason (R): The members of subphylum vertebrata possess notochord during the embryonic period, the notochord is replaced by a cartilaginous or bony vertebral column in adults.

In the light of the above statements, choose the correct answer from the options given below:

Options:

- (a) **A** is true but **R** is false
- (b) **A** is false but **R** is true
- (c) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (d) Both **A** and **R** are true and **R** is **not** the correct explanation of **A**.

Answer: (c)

Solution:

All vertebrates are chordates but all chordates are not vertebrates. This statement is true because all the features of chordates are present in all the vertebrates in some stage of life. In adult vertebrates, the notochord is replaced by the vertebral column and this is not the case for all chordates, hence the assertion is correct and reason is the correct explanation of assertion.

Question 103: What is the main function of the spindle fiber during mitosis?

Options:

- (a) To repair damaged DNA
- (b) To regulate cell growth
- (c) To separate the chromosomes
- (d) To synthesize new DNA

Answer: (c)

Solution:

Spindle fibers help in separating sister chromatids of the chromosome during the anaphase of Mitosis.

Question 104: Match **List-I** with **List-II**.

List-I	List-II
A. Alfred Hershey and Martha	I. <i>Streptococcus pneumoniae</i>
B. Euchromatin	II. Densely packed and dark-stained

C. Frederick Griffith	III. Loosely packed and light-stained
D. Heterochromatin	IV. DNA as genetic material confirmation

Choose the **correct** answer from the options given below:

Options:

- (a) A-IV, B-III, C-I, D-II
- (b) A-III, B-II, C-IV, D-I
- (c) A-II, B-IV, C-I, D-III
- (d) A-IV, B-II, C-I, D-III

Answer: (a)

Solution:

The unequivocal proof that DNA is the genetic material came from the experiments of Alfred Hershey and Martha Chase. Region of chromatin that are loosely packed (and stains light) and are referred to as euchromatin. Frederick Griffith performed Transforming Principle experiments with *Streptococcus pneumoniae*. The chromatin that is more densely packed and stains dark are called as Heterochromatin.

Question 105: Match List-I with List-II.

List-I	List-II
A. Adenosine	I. Nitrogen base
B. Adenylic acid	II. Nucleotide
C. Adenine	III. Nucleoside
D. Alanine	IV. Amino acid

Choose the option with all correct matches.

Options:

- (a) A-III, B-II, C-I, D-IV
- (b) A-II, B-III, C-I, D-IV
- (c) A-III, B-IV, C-II, D-I
- (d) A-III, B-II, C-IV, D-I

Answer: (a)

Solution:

Adenine nitrogenous base joins with ribose sugar to form the nucleoside - Adenosine. Adenosine nucleoside joins with the phosphate group to form nucleotide - Adenylic Acid. Adenine is the purine nitrogenous base. Alanine is the basic amino acid.

Question 106: In frog, the Renal system is a special venous connection that acts to link:

Options:

- (a) Kidney and intestine
- (b) Kidney and lower part of body
- (c) Liver and intestine
- (d) Liver and Kidney

Answer: (b)

Solution:

Renal Portal System is a special vascular connection that connects the kidneys to the lower parts of the body of a frog.

Question 107: Which of the following are the post-transcriptional events in an eukaryotic cell?

- A. Transport of pre-mRNA to cytoplasm prior to splicing
- B. Removing of introns and joining of exons
- C. Addition of methyl group at 5' end of hnRNA.
- D. Addition of adenine residues at 3' end of hnRNA.
- E. Base pairing of two complementary RNAs.

Choose the correct answer from the options given below:

Options:

- (a) B, C, E only
- (b) C, D, E only
- (c) A, B, C only
- (d) B, C, D only

Answer: (d)

Solution:

A) Transport of pre mRNA to cytoplasm doesn't occur before splicing, it occurs after splicing, thus, statement a is incorrect. Statements B, C and D are correct.

Question 108: Polymerase chain reaction (PCR) amplifies DNA following the equation

Options:

- (a) $2n + 1$
- (b) $2N^2$
- (c) N^2
- (d) 2^n

Answer: (d)

Solution:

In PCR, DNA is amplified 2 raised to the power n times.

Question 109: Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Both wind and water pollinated flowers are not very colourful and do not produce nectar.

Reason (R): The flowers produce enormous amount of pollen grains in wind and water pollinated flowers.

In the light of the above statements, choose the correct answer from the options given below:

Options:

- (a) **A** is true but **R** is false
- (b) **A** is false but **R** is true
- (c) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (d) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**

Answer: (d)

Solution:

Both wind and water pollinated flowers are colourless and do not produce nectar, the statement is correct but the reason for this is insects need to be offered rewards in order to bring about pollination and flowers should be attractive, since in both these types of pollinating agents, there isn't a need for colourful petals and nectar, such flowers do not produce nectar and are colourless. The reason statement is correct, but not the correct explanation for the statement.

Question 110: Epiphytes that are growing on a mango branch is an example of which of the following?

Options:

- (a) Predation
- (b) Amensalism
- (c) Commensalism
- (d) Mutualism

Answer: (c)

Solution:

Epiphytes growing on a mango tree are an example of commensalism since the epiphyte is

benefitted by gaining support and access to sunlight, while the mango does not appear to derive any benefit from it.

Question 111: Find the correct statements:

- A. In human pregnancy, the major organ systems are formed at the end of 12 weeks.
- B. In human pregnancy, the major organ systems are formed at the end of 8 weeks.
- C. In human pregnancy heart is formed after one month of gestation.
- D. In human pregnancy, limbs and digits develop by the end of second month.
- E. In human pregnancy the appearance of hair is usually observed in the fifth month.

Choose the correct answer from the options given below:

Options:

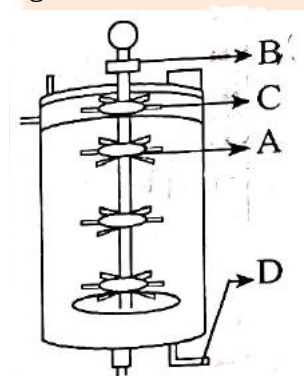
- (a) B, C, D and E only
- (b) A, C, D and E only
- (c) A and E only
- (d) B and C only

Answer: (b)

Solution:

In human pregnancy, the major organ systems are formed at the end of 12 weeks, not 8, thus, statement B is incorrect. Statements A, C, D and E are correct.

Question 112: Identify the part of a bio-reactor which is used as a foam braker from the given figure.



Options:

- (a) D
- (b) C
- (c) A
- (d) B

Answer: (b)

Solution:

Part mentioned as C represents the foam breaker system here.

Question 113: Frogs respire in water by skin and buccal cavity and on land by skin, buccal cavity and lungs.

Choose the correct answer from the following:

Options:

- (a) The statement is false for water and but true for land
- (b) The statement is false for both the environment
- (c) The statement is true for water but false for land
- (d) The statement is true for both the environment

Answer: (a)

Solution:

In water, frogs respire only through skin and not buccal cavity, thus, the statement is wrong, whereas on land it can respire through skin, buccal cavity and lungs, thus, this statement is correct.

Question 114: Consider the following statements regarding function of adrenal medullary hormones:

- A. It causes pupillary constriction
- B. It is a hyperglycemic hormone
- C. It causes piloerection
- D. It increases strength of heart contraction

Choose the correct answer from the options given below:

Options:

- (a) A, C and D Only
- (b) D Only
- (c) C and D Only
- (d) B, C and D Only

Answer: (d)

Solution:

Functions of adrenal medullary hormones- 1) It causes pupillary dilation. 2) It is a hyperglycemic hormone. 3) It causes piloerection. 4) It increases strength of heart contraction. Thus, statement A is incorrect, B, C and D are correct.

Question 115: Read the following statements on plant growth and development.

- A. Parthenocarpy can be induced by auxins.
- B. Plant growth regulators can be involved in promotion as well as inhibition of growth.
- C. Dedifferentiation is a pre-requisite for re-differentiation.
- D. Absciscic acid is a plant growth promoter.
- E. Apical dominance promotes the growth of lateral buds.

Choose the option with all correct statements.

Options:

- (a) A, D, E only
- (b) B, D, E only
- (c) A, B, C only
- (d) A, C, E only

Answer: (c)

Solution:

Absciscic acid is a plant growth inhibitor, not a promoter. Apical dominance prevents the growth of lateral buds, it doesn't promote it. Thus, statements D and E are incorrect and statements A, B and C are correct.

Question 116: Which of the following hormones released from the pituitary is actually synthesized in the hypothalamus?

Options:

- (a) Follicle-stimulating hormone (FSH)
- (b) Adrenocorticotrophic hormone (ACTH)
- (c) Luteinizing hormone (LH)
- (d) Anti-diuretic hormone (ADH)

Answer: (d)

Solution:

Anti Diuretic Hormone (ADH) and Oxytocin are released from the pituitary but synthesized in the hypothalamus. FSH, LH and ACTH are produced and released from the pituitary itself.

Question 117: Which of the following is an example of non-distilled alcoholic beverage produced by yeast?

Options:

- (a) Beer
- (b) Rum
- (c) Whisky
- (d) Brandy

Answer: (a)

Solution:

Beer and wine are produced without distillation, Whisky, Brandy and rum are produced after distillation.

Question 118: What is the pattern of inheritance for polygenic trait?

Options:

- (a) Autosomal dominant pattern
- (b) X-linked recessive inheritance pattern
- (c) Mendelian inheritance pattern
- (d) Non-mendelian inheritance pattern

Answer: (d)

Solution:

Polygenic inheritance requires multiple genes for a single character, thus, this does not fit under any type of Mendelian inheritance, thus, this is Non-Mendelian Inheritance.

Question 119: Match **List-I** and **List-II**.

List-I	List-II
A. Head	I. Enzymes
B. Middle piece	II. Sperm motility
C. Acrosome	III. Energy
D. Tail	IV. Genetic material

Choose the correct answer from the options given below:

Options:

- (a) A-III, B-IV, C-II, D-I
- (b) A-III, B-II, C-I, D-IV
- (c) A-IV, B-III, C-I, D-II
- (d) A-IV, B-III, C-II, D-I

Answer: (c)

Solution:

Head of the sperm contains the genetic material, DNA. Middle piece contains mitochondria, which is responsible for energy. Acrosome contains enzymes which help in dissolution of the membrane of the ovum, tail helps in providing motility to the sperm.

Question 120: Which of the following is an example of a zygomorphic flower?

Options:

- (a) Pea
- (b) Chilli
- (c) *Petunia*
- (d) *Datura*

Answer: (a)

Solution:

Zygomorphic flowers are Pea, Beans, *Cassia* etc., Chilli, *Petunia*, *Datura* are all

actinomorphic flowers.

Question 121: Which of following organisms *cannot* fix nitrogen?

- A. *Azotobacter*
- B. *Oscillatoria*
- C. *Anabaena*
- D. *Volvox*
- E. *Nostoc*

Choose the *correct* answer from the options given below:

Options:

- (a) B only
- (b) E only
- (c) A only
- (d) D only

Answer: (d)

Solution:

Cyanobacteria such as *Anabaena*, *Nostoc* and *Oscillatoria* are N_2 fixing *cyanobacteria*. *Azotobacteria* is a free living N_2 fixing bacteria. *Volvox* is not involved in N_2 fixing.

Question 122: Which one of the following is an example of ex-situ conservation?

Options:

- (a) Zoos and botanical gardens
- (b) Protected areas
- (c) National Park
- (d) Wildlife Sanctuary

Answer: (a)

Solution:

Ex-situ is off site conservation i.e. conservation of living organisms out of their natural habitat. Zoos and Botanical Garden are examples of the same.

Question 123: Who is known as the father of Ecology in India?

Options:

- (a) Ram Udar
- (b) Birbal Sahni
- (c) S. R. Kashyap
- (d) Ramdeo Misra

Answer: (d)

Solution:

Ramdeo Mishra is the father of Indian Ecology

Question 124: Given below are two statements:

Statement I: In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes, RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.

Statement II: DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

Options:

- (a) Statement I is correct but statement II is incorrect

- (b) Statement I is incorrect but statement II is correct
 (c) Both statement I and statement II are correct
 (d) Both statement I and statement II are incorrect

Answer: (c)

Solution:

The first statement correctly explains RNA being the first genetic material as well as being unstable due to its single stranded nature. The second statement is correct as well as DNA evolved from RNA as a more stable double stranded genetic material.

Question 125: Given below are two statements:

Statement I: Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II: RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

Options:

- (a) Statement I is correct but statement II is incorrect
 (b) Statement I is incorrect but statement II is correct
 (c) Both statement I and statement II are correct
 (d) Both statement I and statement II are incorrect

Answer: (b)

Solution:

rRNA and tRNA do interact with mRNA during the process of transcription. RNAi is present in all eukaryotes as a method of cellular defense.

Question 126: Match List-I with List-II.

List-I	List-II
A. Heart	I. Erythropoietin
B. Kidney	II. Aldosterone
C. Gastro-intestinal tract	III. Atrial natriuretic factor
D. Adrenal Cortex	IV. Secretin

Choose the **correct** answer from the options given below:

Options:

- (a) A-I, B-III, C-IV, D-II
 (b) A-III, B-I, C-IV, D-II
 (c) A-II, B-I, C-III, D-IV
 (d) A-IV, B-III, C-II, D-I

Answer: (b)

Solution:

Heart produces ANF, Kidney produces erythropoietin, GI tract produces secretin and Adrenal cortex produces Aldosterone.

Question 127: All living members of the class Cyclostomata are:

Options:

- (a) Symbiotic
 (b) Ectoparasite
 (c) Free living
 (d) Endoparasite

Answer: (b)

Solution:

All living members of class cyclostomata are ectoparasites.

Question 128: Streptokinase produced by *bacterium Streptococcus* is used for

Options:

- (a) Liver disease treatment
- (b) Removing clots from blood vessels
- (c) Curd production
- (d) Ethanol production

Answer: (b)

Solution:

Streptokinase is a clot buster used to remove clots from blood vessels

Question 129: Role of the water vascular system in Echinoderms is:

- A. Respiration and Locomotion
- B. Excretion and Locomotion
- C. Capture and transport of food
- D. Digestion and Respiration
- E. Digestion and Excretion

Choose the **correct** answer from the options given below:

Options:

- (a) B and C Only
- (b) B, D and E Only
- (c) A and B Only
- (d) A and C Only

Answer: (d)

Solution:

Water vascular system is involved in respiration, locomotion, capture of prey and transport of food.

Question 130: Match List I with List II.

List I	List II
A. Pteridophyte	I. <i>Salvia</i>
B. Bryophyte	II. <i>Ginkgo</i>
C. Angiosperm	III. <i>Polytrichum</i>
D. Gymnosperm	IV. <i>Salvinia</i>

Choose the option with all **correct** matches.

Options:

- (a) A-III, B-IV, C-I, D-II
- (b) A-IV, B-III, C-II, D-I
- (c) A-III, B-IV, C-II, D-I
- (d) A-IV, B-III, C-I, D-II

Answer: (d)

Solution:

Salvinia is a pteridophyte, Polytrichum is a Bryophyte, Salvia is a Angiosperms, Ginkgo is a gymnosperms

Question 131: Which are correct:

- A. Computed tomography and magnetic resonance imaging detect cancers of internal organs.
- B. Chemotherapeutics drugs are used to kill non-cancerous cells.
- C. α -interferon activate the cancer patients' immune system and helps in destroying the tumour.
- D. Chemotherapeutic drugs are biological response modifiers.

E. In the case of leukemia blood cell counts are decreases.
Choose the **correct** answer from the options given below:

Options:

- (a) C and D only
- (b) A and C only
- (c) B and D only
- (d) D and E only

Answer: (b)

Solution:

CT and MRI are used to detect cancers of internal organs. α -interferon works by activating the body's immune system against the cancer cells. Chemotherapeutic drugs are not used to kill non-cancerous cells, but target cancerous cells. Chemotherapeutic drugs are not biological response modifiers. In case of leukemia the count of abnormal blood cells increases.

Question 132: What are the potential drawback in adoption of the IVF method?

- A. High fatality risk to mother
- B. Expensive instruments and reagents
- C. Husband/wife necessary for being donors
- D. Less adoption of orphans
- E. Not available in India
- F. Possibility that the early embryo does not survive

Choose the correct answer from the options given below:

Options:

- (a) A, B, C, D only
- (b) A, B, C, E, F only
- (c) B, D, F only
- (d) A, C, D, F only

Answer: (c)

Solution:

Potential drawbacks of IVF are it being an expensive process due to expensive instruments and reagents. Due to IVF less orphans might be adopted and there is a high possibility of the early embryos not surviving.

Question 133: Consider the following:

- A. The reductive division for the human female gametogenesis starts earlier than that of the male gametogenesis.
- B. The gap between the first meiotic division and the second meiotic division is much shorter for males compared to females.
- C. The first polar body is associated with the formation of the primary oocyte.
- D. Luteinizing Hormone (LH) surge leads to disintegration of the endometrium and onset of menstrual bleeding.

Choose the **correct** answer from the options given below:

Options:

- (a) B and D are true
- (b) B and C are true
- (c) A and B are true
- (d) A and C are true

Answer: (c)

Solution:

Oogenesis starts before birth in females while spermatogenesis starts after puberty. In females, the first meiotic division occurs before birth, however is completed during each menstrual cycle followed by the second meiotic division which is again only completed after the entry of sperm but before fertilization. In males the first meiotic division is immediately

followed by the second meiotic division.

Question 134: In bryophytes, the gemmae help in which one of the following?

Options:

- (a) Nutrient absorption
- (b) Gaseous exchange
- (c) Sexual reproduction
- (d) Asexual reproduction

Answer: (d)

Solution:

Asexual reproduction in liverworts takes place by fragmentation of thalli, or by the formation of specialised structures called gemmae (sing. gemma). Gemmae are green, multicellular, asexual buds, which develop in small receptacles called gemma cups located on the thalli. The gemmae become detached from the parent body and germinate to form new individuals.

Question 135: Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The primary function of the Golgi apparatus is to package the materials made by the endoplasmic reticulum and deliver it to intracellular targets and outside the cell.

Reason (R): Vesicles containing materials made by the endoplasmic reticulum fuse with the cis face of the Golgi apparatus, and they are modified and released from the trans face of the Golgi apparatus.

In the light of the above statements, choose the **correct** answer from the options given below:

Options:

- (a) **A** is true but **R** is false
- (b) **A** is false but **R** is true
- (c) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (d) Both **A** and **R** are true but **R** is **not** the correct explanation of **A**

Answer: (c)

Solution:

The Golgi apparatus principally performs the function of packaging materials, to be delivered either to the intra-cellular targets or secreted outside the cell. Materials to be packaged in the form of vesicles from the ER fuse with the cis face of the Golgi apparatus and move towards the maturing face. This explains, why the Golgi apparatus remains in close association with the endoplasmic reticulum.

Question 136: Which one of the following statements refers to Reductionist Biology?

Options:

- (a) Chemical approach to study and understand living organisms.
- (b) Behavioural approach to study and understand living organisms.
- (c) Physico-chemical approach to study and understand living organisms.
- (d) Physiological approach to study and understand living organisms.

Answer: (c)

Solution:

The reductionist approach to study of life forms resulted in increasing use of physico-chemical concepts and techniques. Majority of these studies employed either surviving tissue model or straightaway cell free systems. An explosion of knowledge resulted in molecular biology

Question 137: After maturation, in primary lymphoid organs, the lymphocytes migrate for interaction with antigens to secondary lymphoid organ(s)/tissue(s) like:

- A. thymus
- B. bone marrow
- C. spleen
- D. lymph nodes
- E. Peyer's patches

Choose the *correct* answer from the options given below:

Options:

- (a) E, A, B only
- (b) C, D, E only
- (c) B, C, D only
- (d) A, B, C only

Answer: (b)

Solution:

Secondary lymphoid organs are where lymphocytes mature and interact with antigens to initiate an immune response. These organs include the spleen, lymph nodes, and Peyer's patches.

Primary lymphoid organs are the sites where lymphocytes are produced and mature. These organs are the bone marrow and the thymus. Bone marrow produces B cells, while the thymus matures T cells.

Question 138: Match List I with List II :

	List I		List II
A.	The Evil Quartet	I.	Cryopreservation
B.	Ex situ conservation	II.	Alien species invasion
C.	<i>Lantana camara</i>	III.	Causes of biodiversity losses
D.	Dodo	IV.	Extinction

Choose the option will all correct matches.

Options:

- (a) A-III, B-IV, C-II, D-I
- (b) A-III, B-II, C-IV, D-I
- (c) A-III, B-II, C-I, D-IV
- (d) A-III, B-I, C-II, D-IV

Answer: (d)

Solution:

The correct answer is option (4): A-III, B-I, C-II, D-IV. The Evil Quartet refers to the four major causes of biodiversity loss—habitat loss, overexploitation, alien species invasion, and co-extinction—hence matches with III. Ex situ conservation involves preserving biodiversity outside natural habitats through methods like cryopreservation, matching with I. *Lantana camara* is an invasive alien species that disrupts native ecosystems, so it matches with II. The Dodo is a well-known extinct species, representing extinction, and matches with IV.

Question 139: How many meiotic and mitotic divisions need to occur for the development of a mature female gametophyte from the megaspore mother cell in an angiosperm plant?

Options:

- (a) 1 Meiosis and 3 Mitosis
- (b) No Meiosis and 2 Mitosis
- (c) 2 Meiosis and 3 Mitosis
- (d) 1 Meiosis and 2 Mitosis

Answer: (a)

Solution:

In angiosperms, the process of female gametophyte (embryo sac) development begins with the megaspore mother cell (MMC) inside the ovule:

The MMC undergoes meiosis to form four haploid megaspores. Out of these, only one functional megaspore survives.

So, 1 meiotic division occurs here. The functional megaspore undergoes 3 successive mitotic

divisions:

First mitosis: 2 nuclei

Second mitosis: 4 nuclei

Third mitosis: 8 nuclei

These nuclei then get arranged to form the mature embryo sac (female gametophyte).

Therefore:

1 Meiosis (to form megaspores from MMC)

3 Mitosis (to form 8-nucleate female gametophyte)

Question 140: Which of the following type of immunity is present at the time of birth and is a non-specific type of defence in the human body?

Options:

(a) Cell-mediated Immunity

(b) Humoral Immunity

(c) Acquired Immunity

(d) Innate Immunity

Answer: (d)

Solution:

The correct answer is (d) Innate Immunity. Innate immunity is the non-specific type of defense that is present in the human body from the time of birth. It does not require prior exposure to pathogens and acts as the first line of defense against infections. This immunity includes physical barriers like the skin, physiological barriers like acidic gastric juice, cellular barriers like phagocytic cells, and cytokine barriers that help in fighting infections. In contrast, cell-mediated and humoral immunity are part of acquired immunity, which is specific and develops after birth in response to exposure to antigens. Hence, the type of immunity present at birth and non-specific in nature is innate immunity.

Question 141: Given below are two statements:

Statement I: Fig fruit is a non-vegetarian fruit as it has enclosed fig wasps in it.

Statement II: Fig wasp and fig tree exhibit mutual relationship as fig fruit and fig fruit gets pollinated by fig wasp.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

Options:

(a) Statement I is correct but statement II is incorrect

(b) Statement I is incorrect but statement II is correct

(c) Both statement I and statement II are correct

(d) Both statement I and statement II are incorrect

Answer: (c)

Solution:

Statement I: "Fig fruit is a non-vegetarian fruit as it has enclosed fig wasps in it."

This is biologically correct—many species of figs (genus *Ficus*) have a unique pollination mechanism involving fig wasps. The female wasp enters the fig (which is a syconium, a type of inflorescence) to lay eggs, and in the process, some of them die inside. Thus, technically, some fig fruits may contain dead fig wasps, though they are digested by enzymes and not usually visible when consumed. Hence, some people refer to it as "non-vegetarian," though scientifically it's still classified as a plant product.

Statement II: "Fig wasp and fig tree exhibit mutual relationship as fig wasp completes its life cycle in fig fruit and fig fruit gets pollinated by fig wasp."

This is also correct and well-documented in ecology. The fig tree and fig wasp show obligate mutualism, where both species are dependent on each other for survival and reproduction. The wasp pollinates the fig flowers while laying eggs, and in turn, the fig provides a safe environment for the wasp's larvae to develop.

Question 142: Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.

Reason (R) : Presence of more than one nucleus in the tapetum increases the efficiency of nourishing the developing microspore mother cells.

In light of the above statements, choose the **most appropriate**-answer from the options given below :

Options:

- (a) **A** is true but **R** is false
- (b) **A** is false but **R** is true
- (c) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (d) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**

Answer: (c)

Solution:

Assertion (A): Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.

This is true. The tapetum is the innermost layer of the anther wall and plays a crucial role in nourishing developing pollen grains. Tapetal cells are rich in cytoplasm and often become binucleate or even multinucleate, due to either mitotic divisions without cytokinesis or nuclear fusion.

Reason (R): Presence of more than one nucleus in the tapetum increases the efficiency of nourishing the developing microspore mother cells.

This is also true and correctly explains the assertion. The multinucleate condition leads to high metabolic activity, allowing tapetal cells to synthesize and transport large amounts of nutrients, enzymes, and sporopollenin precursors essential for pollen development.

Question 143: From the statements given below choose the correct option :

- A. The eukaryotic ribosomes are 80S and prokaryotic ribosomes are 70S.
- B. Each ribosome has two sub-units.
- C. The two sub-units of 80S ribosome are 60S and 40S while that of 70S are 50S and 30S.
- D. The two sub-units of 80S ribosome are 60S and 20S and that of 70S are 50S and 20S.
- E. The two sub-units of 80S are 60S and 30S and that of 70S are 50S and 30S.

Options:

- (a) A, B, E are true
- (b) B, D, E are true
- (c) A, B, C are true
- (d) A, B, D are true

Answer: (c)

Solution:

Ribosomes are non-membranous organelles involved in protein synthesis and are found in both prokaryotic and eukaryotic cells. Statement A is correct because eukaryotic ribosomes are 80S, and prokaryotic ribosomes are 70S in size, where 'S' stands for Svedberg unit, indicating sedimentation rate during centrifugation.

Statement B is also correct, as each ribosome is composed of two subunits. Statement C is correct because the 80S ribosome consists of a 60S (large) and 40S (small) subunit, and the 70S ribosome consists of a 50S (large) and 30S (small) subunit. Statements D and E are incorrect as they mention wrong subunit combinations that are not found in nature. Therefore, the correct combination of true statements is A, B, and C.

Question 144: Which one of the following enzymes contains 'Haem' as the prosthetic group?

Options:

- (a) Succinate dehydrogenase
- (b) Catalase
- (c) RuBisCo

(d) Carbonic anhydrase

Answer: (b)

Solution:

Catalase: This enzyme does contain a heme group

Question 145: What is the name of the blood vessel that carries deoxygenated blood from the body to the heart in a frog ?

Options:

- (a) Pulmonary vein
- (b) Vena cava
- (c) Aorta
- (d) Pulmonary artery

Answer: (b)

Solution:

In a frog, the vena cava is the blood vessel that carries deoxygenated blood from the body to the heart. The deoxygenated blood from various parts of the body enters the right atrium of the heart through the anterior and posterior vena cava.

Question 146: Given below are the stages in the life cycle of pteridophytes. Arrange the following stages in the correct sequence.

- A. Prothallus stage
- B. Meiosis in spore mother cells
- C. Fertilisation
- D. Formation of archegonia and antheridia in gametophyte.
- E. Transfer of antherozoids to the archegonia in presence of water.

Choose the **correct** answer from the options given below :

Options:

- (a) D, E, C, A, B
- (b) E, D, C, B, A
- (c) B, A, D, E, C
- (d) B, A, E, C, D

Answer: (c)

Solution:

Let's go step-by-step through the life cycle of pteridophytes:

B. Meiosis in spore mother cells occurs in the sporangia of the sporophyte to produce haploid spores.

A. Prothallus stage: These spores germinate to form the gametophyte called prothallus.

D. Formation of archegonia and antheridia in gametophyte

The mature prothallus develops sex organs: antheridia (male) and archegonia (female).

E. Transfer of antherozoids to the archegonia in presence of water

Water is necessary for the motile antherozoids (sperm) to swim to the archegonia.

C. Fertilisation occurs when the antherozoid fuses with the egg in the archegonium, forming a diploid zygote, which develops into a new sporophyte.

Question 147: The blue and white selectable markers have been developed which differentiate recombinant colonies from non-recombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate.

Given below are two statements about this method:

Statement I : The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.

Statement II : The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

Options:

- (a) Statement I is correct but Statement II is incorrect
- (b) Statement I is incorrect but Statement II is correct
- (c) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

Answer: (b)

Solution:

Statement I: The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.

Incorrect. In blue-white screening, non-recombinant colonies (those without the gene insert) turn blue because they can produce a functional β -galactosidase enzyme, which breaks down the chromogenic substrate (X-gal) and produces the blue color. The presence of an insert in the plasmid disrupts the β -galactosidase gene, leading to a white color.

Statement II: The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.

Correct. The white colonies are recombinant because the insertion of foreign DNA disrupts the β -galactosidase gene, preventing the formation of the blue color.

Question 148: Which of the following microbes is **NOT** involved in the preparation of household products?

- A. *Aspergillus niger*
- B. *Lactobacillus*
- C. *Trichoderma polysporum*
- D. *Saccharomyces cerevisiae*
- E. *Propionibacterium sharmanii*

Choose the **correct** answer from the options given below:

Options:

- (a) C and D only
- (b) C and E only
- (c) A and B only
- (d) A and C only

Answer: (d)

Solution:

Aspergillus niger and *Trichoderma polysporum* are not involved in preparation of household products.

Question 149: Silencing of specific mRNA is possible via RNAi because of -

Options:

- (a) Complementary tRNA
- (b) Non-complementary ssRNA
- (c) Complementary dsRNA
- (d) Inhibitory ssRNA

Answer: (c)

Solution:

RNA interference (RNAi) is a gene-silencing mechanism used by cells to regulate gene expression or defend against viral RNA. It works through double-stranded RNA (dsRNA) that is complementary to the target mRNA.

This is how it works:

Introduction of complementary dsRNA into the cell.

The dsRNA is recognized and cleaved by an enzyme called Dicer into small fragments called siRNA (small interfering RNA).

These siRNAs bind to a protein complex called RISC (RNA-induced silencing complex).

The siRNA guides RISC to the complementary mRNA, leading to its cleavage and silencing (i.e., preventing translation).

Therefore, complementary double-stranded RNA (dsRNA) is crucial for initiating RNAi.

Question 150: The complex II of mitochondrial electron transport chain is also known as

Options:

- (a) Cytochrome c oxidase
- (b) NADH dehydrogenase
- (c) Cytochrome bc₁
- (d) Succinate dehydrogenase

Answer: (d)

Solution:

Complex II- Succinate dehydrogenase:

This enzyme is responsible for oxidizing succinate to fumarate in the citric acid cycle. It also plays a crucial role in the electron transport chain by transferring electrons from succinate to ubiquinone (Q).

Question 151: While trying to find out the characteristic of a newly found animal, a researcher did the histology of adult animal and observed a cavity with presence of mesodermal tissue towards the body wall but no mesodermal tissue was observed towards the alimentary canal. What could be the possible coelom of that animal?

Options:

- (a) Schizocoelomate
- (b) Spongocoelomate
- (c) Acoelomate
- (d) Pseudocoelomate

Answer: (d)

Solution:

The above explanation defines pseudocoelom condition

Question 152: Given below are two statements :

Statement I : In a floral formula $\oplus$ stands for zygomorphic nature of the flower, and $\underline{G}$ stands for inferior ovary.

Statement II : In a floral formula $\oplus$ stands for actinomorphic nature of the flower and $\underline{G}$ stands for superior ovary.

In the light of the above statements, choose the **correct** answer from the options given below:

Options:

- (a) Statement I is correct but Statement II is incorrect
- (b) Statement I is incorrect but Statement II is correct
- (c) Both Statement I and Statement II are correct
- (d) Both Statement I and Statement II are incorrect

Answer: (b)

Solution:

Statement II is correct

Question 153: Given below are two statements :

Statement I : In ecosystem, there is unidirectional flow of energy of sun from producers to consumers.

Statement II : Ecosystems are exempted from 2nd law of thermodynamics.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

Options:

- (a) Statement I is correct but statement II is incorrect
- (b) Statement I is incorrect but statement II is correct
- (c) Both statement I and statement II are correct
- (d) Both statement I and statement II are incorrect

Answer: (a)

Solution:

Ecosystems are not exempted from the second law of thermodynamics

Question 154: Which of the following is the unit of productivity of an Ecosystem?

Options:

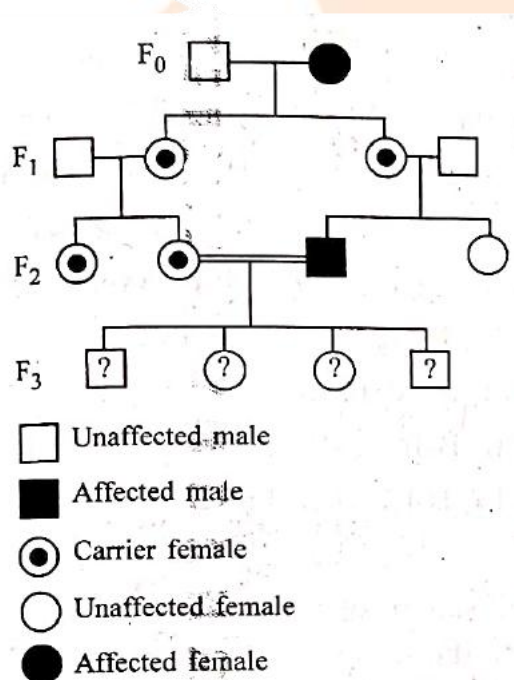
- (a) KCal m^{-3}
- (b) $(\text{KCal m}^{-2})\text{yr}^{-1}$
- (c) gm^{-2}
- (d) KCal m^{-2}

Answer: (b)

Solution:

The rate of biomass production is called productivity. It is expressed in terms of $\text{gm}^{-2} \text{yr}^{-1}$ or $(\text{kcal m}^{-2}) \text{yr}^{-1}$

Question 155: With the help of given pedigree, find out the probability for the birth of a child having no disease and being a carrier (has the disease mutation in one allele of the gene) in F_3 generation.



Options:

- (a) $1/8$
- (b) Zero
- (c) $1/4$
- (d) $1/2$

Answer: (c) or (d)

Solution:

Both (c) and (d) are possible answers.

Question 156: In the seeds of cereals, the outer covering of endosperm separates the embryo by a protein-rich layer called :

Options:

- (a) Integument
- (b) Aleurone layer
- (c) Coleoptile
- (d) Coleorhiza

Answer: (b)

Solution:

In seeds of cereals the outer covering of endosperm separates the embryo by a protein rich layer called Aleurone layer

Question 157: Match List I with List II:

	List I		List II
A.	Chlorophyll a	I.	Yellow-green
B.	Chlorophyll b	II.	Yellow
C.	Xanthophylls	III.	Blue-green
D.	Carotenoids	IV.	Yellow to Yellow-orange

Choose the option with all **correct** matches.

Options:

- (a) A-I, B-II, C-IV, D-III
- (b) A-I, B-IV, C-III, D-II
- (c) A-III, B-IV, C-II, D-I
- (d) A-III, B-I, C-II, D-IV

Answer: (d)

Solution:

Chlorophyll a - Blue green, Chlorophyll b - Yellow green, Xanthophyll - yellow, Carotenoids - Yellow to yellow orange

Question 158: Who proposed that the genetic code for amino acids should be made up of three nucleotides?

Options:

- (a) Jacques Monod
- (b) Franklin Stahl
- (c) George Gamow
- (d) Francis Crick

Answer: (c)

Solution:

George Gamow proposed that the genetic code for amino acids should be made up of three nucleotides

Question 159: Histones are enriched with -

Options:

- (a) Phenylalanine & Leucine
- (b) Phenylalanine & Arginine
- (c) Lysine & Arginine
- (d) Leucine & Lysine

Answer: (c)

Solution:

Histones are enriched with Lysine and Arginine

Question 160: Which of the following enzyme(s) are NOT essential for gene cloning?

- A. Restriction enzymes
- B. DNA ligase
- C. DNA mutase
- D. DNA recombinase
- E. DNA polymerase

Choose the **correct** answer from the options given below :

Options:

- (a) D and E only
- (b) B and C only
- (c) C and D only
- (d) A and B only

Answer: (c)

Solution:

Restriction enzymes, DNA Ligase and DNA polymerase are essential for gene cloning

Question 161: A specialised membranous structure in a prokaryotic cell which helps in cell wall formation, DNA replication and respiration is :

Options:

- (a) Cristae
- (b) Endoplasmic Reticulum
- (c) Mesosome
- (d) Chromatophores

Answer: (c)

Solution:

A specialised membranous structure in a prokaryotic cell which helps in cell wall formation, DNA replication and respiration is mesosome

Question 162: Which factor is important for termination of transcription?

Options:

- (a) ρ (rho)
- (b) γ (gamma)
- (c) α (alpha)
- (d) σ (sigma)

Answer: (a)

Solution:

ρ (rho) factor is important for termination of transcription

Question 163: Which of the following statement is **correct** about location of the male frog copulatory pad ?

Options:

- (a) Second digit of fore limb
- (b) First digit of the fore limb
- (c) First and Second digit of fore limb
- (d) First digit of hind limb

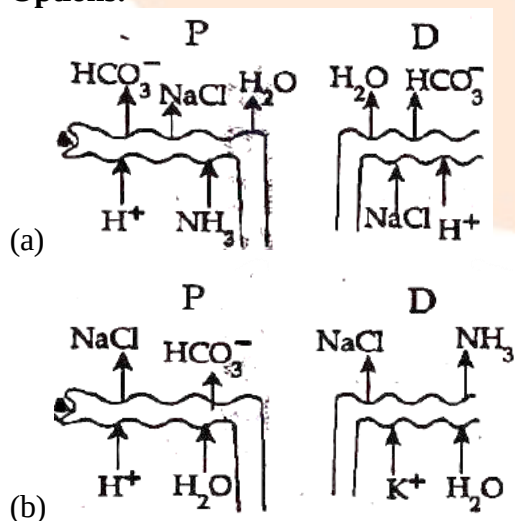
Answer: (b)

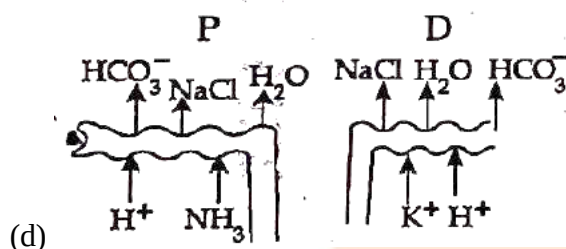
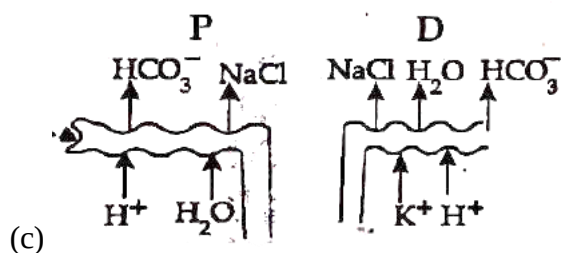
Solution:

In male frog copulatory pad is present in first digit of the fore limb

Question 164: Which of the following diagrams is correct with regard to the proximal (P) and distal (D) tubule of the Nephron.

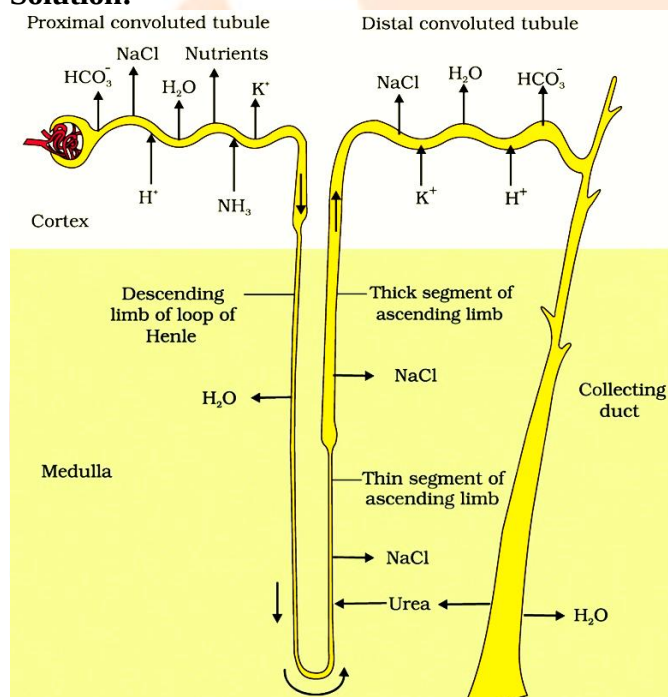
Options:





Answer: (d)

Solution:



Question 165: Identify the statement that is NOT correct.

Options:

- (a) Antigen binding site is located at C-terminal region of antibody molecules.
- (b) Constant region of heavy and light chains are located at C-terminus of antibody molecules.
- (3) Each antibody has two light and two heavy chains.
- (4) The heavy and light chains are held together by disulfide bonds.

Answer: (a)

Solution:

Antigen binding site is located at N-terminal region of antibody molecules.

Question 166: Match List I with List II :

	List I		List II
A.	Scutellum	I.	Persistent nucellus
B.	Non-albuminous seed	II.	Cotyledon of Monocot seed
C.	Epiblast	III.	Groundnut
D.	Perisperm	IV.	Rudimentary cotyledon

Choose the option with all **correct** matches

Options:

- (a) A-IV, B-III, C-I, D-II
- (b) A-II, B-IV, C-III, D-I
- (c) A-II, B-III, C-IV, D-I
- (d) A-IV, B-III, C-II, D-I

Answer: (c)**Solution:**

Scutellum means - In the grass family (monocot) the shield shaped cotyledon is called scutellum that is situated towards one side (lateral) of the embryonal axis.

Non-albuminous seeds, also known as **ex-albuminous or non-endospermic seeds**, are seeds where the endosperm is completely consumed by the developing embryo before the seed matures,

Epiblast means - In monocotyledonous plants (like grasses), the epiblast is indeed considered a rudimentary cotyledon,

Perisperm - Occasionally, remnants of nucellus are also persistent. This residual, persistent nucellus is the perisperm. It's a diploid tissue that provides nourishment to the developing embryo and is found in seeds like black pepper and beet

Question 167: Find the statement that is **NOT** correct with regard to the structure of monocot stem.

Options:

- (a) Vascular bundles are conjoint and closed
- (b) Phloem parenchyma is absent.
- (c) Hypodermis is parenchymatous.
- (d) Vascular bundles are scattered

Answer: (c)**Solution:**

The monocot stem has a **sclerenchymatous hypodermis**, a large number of **scattered vascular bundles**, Vascular bundles are **conjoint and closed & phloem parenchyma is absent**.

Question 168: Twins are born to a family that lives next door to you. The twins are a boy and a girl. Which of the following must be true?

Options:

- (a) They were conceived through in vitro fertilization.
- (b) They have 75% identical genetic content.
- (c) They are monozygotic twins.
- (d) They are fraternal twins.

Answer: (d)**Solution:**

Fraternal twins, also known as dizygotic twins, occur when two separate eggs are fertilized by two different sperm. Since a boy and a girl are born, it indicates two separate eggs and sperm were involved, meaning they are fraternal twins. Monozygotic twins result from one fertilized egg splitting, leading to identical genetic content and the same sex. In vitro fertilization is a method of assisted reproduction but doesn't determine whether twins will be fraternal or identical.

Question 169: Sweet potato and potato represent a certain type of evolution. Select the correct combination of terms to explain the evolution.

Options:

- (a) Homology, convergent
- (b) Analogy, divergent
- (c) Analogy, convergent
- (d) Homology, divergent

Answer: (c)

Solution:

Sweet potato (root modification) and **potato (stem modification)** is an example for **analogy and convergent evolution**, as they have different origins but perform the same function (food storage)

Question 170: Which one of the following phytohormones promotes nutrient mobilization which helps in the delay of leaf senescence in plants?

Options:

- (a) Gibberellin
- (b) Cytokinin
- (c) Ethylene
- (d) Absciscic acid

Answer: (b)

Solution:

Cytokinin promote nutrient mobilisation which helps in the delay of leaf senescence.

Question 171: Why can't insulin be given orally to diabetic patients?

Options:

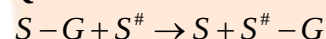
- (a) Because of structural variation
- (b) Its bioavailability will be increased
- (c) Human body will elicit strong immune response
- (d) It will be digested in Gastro-Intestinal (GI) tract

Answer: (d)

Solution:

Insulin cannot be taken orally by diabetic people because it is a protein hormone that is broken down by digestive **enzymes in the stomach and intestines**.

Question 172: Name the class of enzyme that usually catalyze the following reaction:



Where, G $\rightarrow$ a group other than hydrogen

S $\rightarrow$ a substrate

S[#] $\rightarrow$ another substrate

Options:

- (a) Transferase
- (b) Ligase
- (c) Hydrolase
- (d) Lyase

Answer: (a)

Solution:

Transferases: Enzymes catalysing a transfer of a group, G (other than hydrogen) between a pair of substrate S and S'



Where, G - a group other than hydrogen, S - a substrate & S' - another substrate

Question 173: Given below are two statements :

Statement I : The DNA fragments extracted from gel electrophoresis can be used in construction of recombinant DNA.

Statement II : Smaller size DNA fragments are observed near anode while larger fragments are found near the wells in an agarose gel.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

Options:

- (a) Statement I is correct but statement II is incorrect

- (b) Statement I is incorrect but statement II is correct
 (c) Both statement I and statement II are correct
 (d) Both statement I and statement II are incorrect

Answer: (c)

Solution:

Statement I - The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as **elution**. The DNA fragments purified in this way are used in **constructing recombinant DNA** by joining them with cloning vectors.

Statement II - The DNA fragments separate (resolve) according to their size through sieving effect provided by the agarose gel. Hence, the smaller the fragment size, the farther it moves towards anode while **larger fragments** are move slowly so found **near wells in an agrose gel**.

Question 174: The correct sequence of events in the life cycle of bryophytes is

- Fusion of antherozoid with egg.
- Attachment of gametophyte to substratum.
- Reduction division, to produce haploid spores.
- Formation of sporophyte.
- Release of antherozoids into water.

Choose the **correct** answer from the options given below :

Options:

- B, E, A, D, C
- D, E, A, B, C
- D, E, A, C, B
- B, E, A, C, D

Answer: (a)

Solution:

- The **plant body of bryophytes** is thallus-like and prostrate or erect, and **attached to the substratum** by unicellular or multicellular rhizoids.
- The male sex organ is called antheridium. They produce **biflagellate antherozoids**. The **antherozoids are released into water** where they come in contact with archegonium.
- An antherozoid **fuses with the egg to produce the zygote**.
- Zygotes do not undergo reduction division immediately. They produce a **multicellular body called a sporophyte**.
- Some cells of the sporophyte** undergo reduction division (meiosis) **to produce haploid spores**. These spores germinate to produce gametophyte.

Question 175: Genes R and Y follow independent assortment. If RRYYY produce round yellow seeds and rryy produce wrinkled green seeds, what will be the phenotypic ratio of the F₂ generation?

Options:

- Phenotypic ratio - 9 : 3 : 3 : 1
- Phenotypic ratio - 9 : 7
- Phenotypic ratio - 1 : 2 : 1
- Phenotypic ratio - 3 : 1

Answer: (a)

Solution:

Results of a dihybrid cross - Phenotypic ratio of F₂ generation is

9 Round, Yellow

3 Wrinkled, Yellow

3 Round, Green

1 Wrinkled, Green

Question 176: Each of the following characteristics represent a Kingdom proposed by Whittaker. Arrange the following in increasing order of complexity of body organization.

- A. Multicellular heterotrophs with cell wall made of chitin.
 B. Heterotrophs with tissue/organ/organ system level of body organization.
 C. Prokaryotes with cell wall made of polysaccharides and amino acids.
 D. Eukaryotic autotrophs with tissue/organ level of body organization.
 E. Eukaryotes with cellular body organization.

Choose the **correct** answer from the options given below :

Options:

- (a) A, C, E, D, B
 (b) C, E, A, B, D
 (c) A, C, E, B, D
 (d) C, E, A, D, B

Answer: (d)

Solution:

According to the R.H. Whittaker (1969)

1. **Monera** - Prokaryotic cell which cell wall made up of Non-cellulosic (Polysaccharide + amino acid)
2. **Protista** - Eukaryotic with cellular body organisation
3. **Fungi** - Eukaryotic multicellular heterotrophs with cell wall made of chitin
4. **Plantae** - Eukaryotic autotrophs with tissue/ organ level of organisation.
5. **Animalia** - Heterotrophs with Tissue/organ/ organ system level of organisation.

Question 177: Match List - I with List - II.

	List – I		List - II
A.	Centromere	I.	Mitochondrion
B.	Cilium	II.	Cell division
C.	Cristae	III.	Cell movement
D.	Cell membrane	IV.	Phospholipid Bilayer

Choose the **correct** answer from the options given below :

Options:

- (a) A-IV, B-II, C-III, D-I
 (b) A-II, B-III, C-I, D-IV
 (c) A-I, B-II, C-III, D-IV
 (d) A-II, B-I, C-IV, D-III

Answer: (b)

Solution:

Primary constriction or the centromere on the sides of which disc shaped structures called kinetochores are present. Centromere holds two chromatids of a chromosome. Centromere involved in the cell division during the splitting of sister chromatids.

Cilia (sing.: cilium) and flagella (sing.: flagellum) are hair-like outgrowths of the cell membrane. Cilia are small structures which work like oars, causing the movement of either the cell or the surrounding fluid.

The inner membrane forms a number of infoldings called the **cristae** (sing.: crista) towards the matrix. The cristae increase the surface area.

Cell Membrane - The major lipids are phospholipids that are arranged in a bilayer.

Question 178: Which one of the following equations represents the Verhulst-Pearl Logistic Growth of population?

Options:

- (a) $\frac{dN}{dt} = rN \left(\frac{N - K}{N} \right)$
 (b) $\frac{dN}{dt} = N \left(\frac{r - K}{K} \right)$
 (c) $\frac{dN}{dt} = r \left(\frac{K - N}{K} \right)$

$$(d) \frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$$

Answer: (d)

Solution:

Verhulst-Pearl Logistic Growth equation is $N/dt = rN (K - N/K)$ -

Question 179: Match **List - I** with **List - II**

	List - I		List - II
A.	Emphysema	I.	Rapid spasms in muscle due to low Ca^{++} in body fluid
B.	Angina Pectoris	II.	Damaged alveolar walls and decreased respiratory surface
C.	Glomerulo-nephritis	III.	Acute chest pain when not enough oxygen is reaching to heart muscle
D.	Tetany	IV.	Inflammation of glomeruli of kidney

Choose the **correct** answer from the options given below :

Options:

(a) A-II, B-IV, C-III, D-I

(b) A-II, B-III, C-IV, D-I

(c) A-III, B-I, C-IV, D-II

(d) A-III, B-I, C-II, D-IV

Answer: (b)

Solution:

Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased. One of the major causes of this is cigarette smoking

Angina: It is also called 'angina pectoris'. A symptom of acute chest pain appears when not enough oxygen is reaching the heart muscle. Angina can occur in men and women of any age but it is more common among the middle-aged and elderly. It occurs due to conditions that affect the blood flow.

Glomerulonephritis: Inflammation of glomeruli of kidney,

Tetany: Rapid spasms (wild contractions) in muscle due to low Ca^{++} in body fluid.

Question 180: Cardiac activities of the heart are regulated by :

A. Nodal tissue

B. A special neural centre in the medulla oblongata

C. Adrenal medullary hormones

D. Adrenal cortical hormones

Choose the **correct** answer from the options given below :

Options:

(a) A, C and D Only

(b) A, B and D Only

(c) A, B and C Only

(d) A, B, C and D

Answer: (c)

Solution:

Heart are regulated intrinsically, i.e., auto regulated by specialised muscles (**nodal tissue**),

A special neural centre in the medulla oblongata can moderate the cardiac function through autonomic nervous system (ANS), **Adrenal medullary** hormones can also increase the cardiac output.

(the **adrenal cortex**, which produces cortisol and aldosterone, does not directly increase heart rate in the same way. These hormones from the adrenal cortex primarily regulate blood pressure and blood volume, indirectly impacting the heart.)